

# Hierarchical Mesh Representation Learning With Spectral Dictionary Embedding

Zhongpai Gao<sup>✉</sup>, *Member, IEEE*, Junchi Yan<sup>✉</sup>, *Senior Member, IEEE*, Tianyu Luan<sup>✉</sup>,  
Guangtao Zhai<sup>✉</sup>, *Fellow, IEEE*, and Xiaokang Yang<sup>✉</sup>, *Fellow, IEEE*

**Abstract**—Learning mesh representation is important for many 3D tasks. Conventional convolution for regular data (i.e., images) cannot directly be applied to meshes since each vertex's neighbors are unordered. Previous methods use isotropic filters or predefined local coordinate systems or learning weighting matrices for each template vertex to overcome the irregularity. Learning weighting matrices to resample the vertex's neighbors into an implicit canonical order is the most effective way to capture the local structure of each vertex. However, learning weighting matrices for each vertex increases the model size linearly with the vertex number. Thus, large parameters are required for high-resolution 3D shapes, which is not favorable for many applications. In this paper, we learn spectral dictionary (i.e., bases) for the weighting matrices such that the model size is independent of the resolution of 3D shapes. The coefficients of the weighting matrix bases are learned from the spectral features of the template and its hierarchical levels in a weight-sharing manner. Furthermore, we introduce an adaptive sampling method that learns the hierarchical mapping matrices directly to improve the performance without increasing the model size at the inference stage. Comprehensive experiments demonstrate that our model produces state-of-the-art results with a much smaller model size.

**Index Terms**—Graph convolutional network, representation learning, 3D mesh, 3D morphable model.

## I. INTRODUCTION

RECENTLY, convolutional operations tailored for 3D mesh processing in deep neural networks have garnered significant interest. These operations are pivotal for efficiently and effectively learning representations of meshes, facilitating advancements in a broad array of 3D tasks such as reconstruction [1], [2], [3], shape correspondence [4], [5], synthesis and modeling [6], face recognition [7], segmentation [8], [9], and applications in computer graphics like creating virtual avatars [10].

Received 20 March 2024; revised 3 February 2026; accepted 25 April 2026. Date of publication 4 May 2026; date of current version 6 August 2026. This work was supported by the Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China under Grant JYB2025XD411. Recommended for acceptance by I.A. Kakadiaris. (Corresponding authors: Zhongpai Gao; Junchi Yan.)

Zhongpai Gao is with United Imaging Intelligence, Boston, MA 01803 USA (e-mail: gaozhongpai@gmail.com).

Junchi Yan is with the School of Artificial Intelligence, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: yanjunchi@sjtu.edu.cn).

Tianyu Luan is with the Computer Science & Engineering Department, State University of New York at Buffalo, Buffalo, NY 14228 USA (e-mail: tianyulu@buffalo.edu).

Guangtao Zhai and Xiaokang Yang are with the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: zhaiguangtao@sjtu.edu.cn; xkyang@sjtu.edu.cn).

Digital Object Identifier 10.1109/TPAMI.2026.3690051

The proven efficacy of convolutional neural networks (CNNs) with euclidean-structured data (e.g., images, audio, and computed tomography scans) has led researchers to explore CNNs' applicability to 3D meshes. Unlike their euclidean counterparts, 3D meshes exhibit non-euclidean structure, typically represented as graphs with varying numbers and orientations of neighboring vertices (vertex inconsistency). Hence, developing a convolutional operation that can adapt to the unique features of fixed-topology meshes, akin to CNNs, is crucial for mesh representation learning.

The development of graph convolutional networks (GCNs) has significantly advanced the handling of 3D meshes. Deferrard et al. [20] introduced spectral filters leveraging Chebyshev polynomials, known as *ChebNet*, to offer isotropic filtering based on spectral graph theory. Following this, Ranjan et al. [13] developed a convolutional mesh autoencoder (COMA) specifically for fixed-topology 3D meshes, building on ChebNet. They also implemented a mesh sampling operation to facilitate a hierarchical representation, thereby capturing the non-linear variations observed in human faces more effectively. However, ChebNet's reliance on isotropic rather than anisotropic filters has been critiqued for limiting its expressiveness, particularly when compared to conventional CNNs. Addressing this limitation, Bouritsas et al. [14] proposed *SpiralNet*, employing anisotropic filters in graph convolutions by introducing a novel spiral convolution operator. It attempts to tackle the inconsistency in local structure across different vertices. Nonetheless, as shown in recent findings (Table II of [12]), employing a randomized neighbor order can, intriguingly, yield superior performance, suggesting that pre-defined spiral ordering may not be essential for effective mesh processing.

In [12], a new convolutional operation is introduced, termed *LSA-Conv*, specifically designed to be aware of the local structure around each vertex in a mesh. This operation dynamically learns weighting matrices tailored to the unique neighborhood structure of each vertex, effectively organizing the vertex's neighbors into a pseudo-canonical order. This organization allows for the application of conventional convolution using anisotropic filters, thereby enhancing the model's ability to capture complex geometric details. However, a significant drawback of this approach is the treatment of the weighting matrices as model parameters, which are directly learned during training. Consequently, the size of the model scales linearly with the number of vertices in the template, leading to potentially unwieldy models for high-resolution 3D shapes, especially in

light of advancements in depth sensors that capture dense point clouds with intricate geometric detail. Although a matrix factorization technique was proposed to mitigate this issue, the model size still increases in direct proportion to the shape's resolution. Therefore, while LSA-Conv marks a significant advancement in local structure-aware mesh processing, its scalability is limited by the inherent increase in model size for applications involving high-resolution 3D shapes.

We introduce a novel convolutional approach named **Spectral Dictionary-based Convolution (SDConv)**, which adopts a distributed representation strategy for each mesh vertex. Inspired by dictionary learning, this method involves the construction of a compact set of weighting matrix bases, forming a spectral dictionary. These bases are then linearly combined to generate unique weighting matrices for individual vertices, effectively enabling the adaptation of conventional CNNs with anisotropic filters to meshes by reordering each vertex's neighbors into a pseudo-canonical arrangement. The coefficients that determine the combination of bases for each vertex are derived from balanced spectral features of the mesh template and its hierarchical levels (Fig. 2), ensuring that the model's complexity does not scale with the mesh resolution. Additionally, SDConv incorporates a skip connection, implemented as a linear layer that aligns the dimensions of the output features, to enhance the learning capability.

In mesh representation learning, the significance of sampling operations parallels that of convolutional processes, facilitating the aggregation of mesh features across varying scales within a neural network. Traditional approaches [12], [13], [14] rely on preprocessing stages for feature aggregation across hierarchical levels, adhering to surface error approximations through quadric metrics. This static sampling strategy, however, can constrain the neural network's capacity for representation. Drawing inspiration from Zhang et al. [21], who introduced an attention-based module for learning mapping matrices to enhance feature aggregation, we propose an innovative feature aggregation approach. Our method dynamically learns the mapping matrices, thereby enabling the adjustment of hierarchical levels directly, as shown in Fig. 2(a). This adaptability allows the spectral features of the mesh's hierarchical structures to evolve during training, unlocking further capabilities of SDConv, which we extend to adaptive hierarchical SDConv (HSDConv). These adaptable hierarchical mapping matrices serve as a versatile, train-only drop-in substitute for the static preprocessing matrices in existing architectures, without contributing to model size during inference.

Aligned with the advancements in LSA-Conv [12], COMA [13], SpiralNet [14], SpiralNet++ [15], VCMeshConv [17], Adaptive-Spiral [16], and SIS [22], our SDConv framework is specifically crafted for meshes with fixed topology. Our evaluation framework encompasses two core benchmarks: autoencoder-based reconstruction and shape correspondence, underscoring the adaptability of SDConv and its enhanced variant, HSDConv, in managing fixed topology mesh data. Note that meshes with arbitrary topologies can directly be handled by point-cloud based methods [18], [23]. In this realm, we demonstrate that our approach not only rivals but sometimes exceeds the capabilities of LSA-Conv, especially in achieving a more compact model size. This is evidenced across two

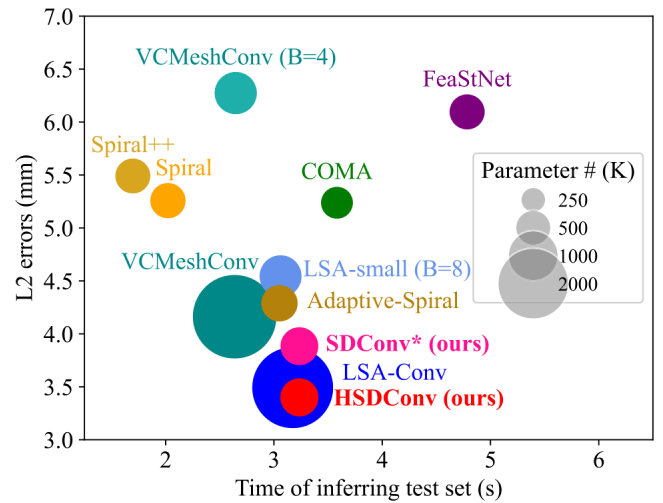


Fig. 1. Quantitative evaluation of our SDConv\* [11] and hierarchical SDConv (HSDConv) against LSA-Conv and LSA-small [12], COMA [13], Spiral [14], SpiralNet++ [15], Adaptive-Spiral [16], VCMeshConv [17], and FeaStNet [18] on the DFAUST dataset [19] for latent size  $d = 32$  with the same network architecture, in terms of reconstruction errors, inference time complexity, and model size. LSA-small is a parameter-reduction version of LSA-Conv. Note, HSDConv outperforms LSA-Conv with a smaller model size. The times are measured on an NVIDIA RTX2080Ti GPU. We repeat each experiment 100 times to reduce measurement variance.

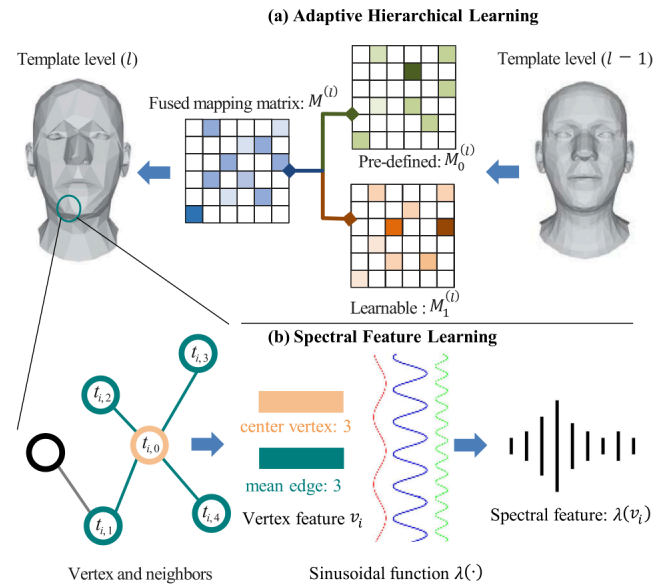


Fig. 2. Extended contributions over the conference paper [11]: (a) an adaptive sampling technique to refine hierarchical levels and (b) spectral feature learning from a balanced fusion of the center vertex and its neighbors.

specialized 3D shape datasets—human faces (COMA [13]) and human bodies (DFAUST [19]), as illustrated in Fig. 1. Additionally, we extend the applicability of our methodology to the domain of monocular 3D hand reconstruction, utilizing the FreiHAND [24] dataset to further validate the efficacy and robustness of our proposed solutions.

Our approach synergizes the strengths of both global and local mesh processing methodologies. Unlike global strategies such as COMA [13], which employ isotropic filters, or SpiralNet [14], which utilizes a uniform model across vertices with predefined

local coordinates, SDConv offers a tailored vertex representation model. This model grants enhanced local adaptability by personalizing the processing for each vertex. Moreover, SDConv presents a significant improvement over other local structure-aware methods like those in [12], which individually learn weighting matrices for each vertex, thereby escalating the model size. SDConv achieves a more streamlined model footprint through a sophisticated dictionary-based encoding strategy, effectively minimizing the model size without compromising performance (refer to Fig. 1 for comparative analysis). Additionally, diverging from the predetermined sampling strategies seen in COMA [13], SpiralNet [14], and LSA-Conv [12] that rely on preprocessed mapping matrices, our method HSDConv innovatively learns the mapping matrices. This learning facilitates dynamic hierarchical feature aggregation, further enhancing our model's capability to capture and represent complex mesh structures efficiently.

In summary, our contributions are as follows:

- 1) We propose to learn distributed representation of each mesh vertex, encoded by a spectral dictionary which is embodied as an inherent component in our carefully devised convolutional operation (SDConv) for 3D meshes. SDConv constructs weighting matrices for each vertex from a small number of learnable bases to resample its neighbors based on the local neighboring structure.
- 2) For more expressive vertex feature learning, especially for high-frequency features, we use spectral features with high-frequency information preserved to learn the corresponding coefficients of weighting matrix bases. Furthermore, an adaptive temperature for softmax is introduced to control the 'softness' of the output distribution of the basis coefficients. Also, we introduce a skip connection to SDConv, which further improves the performance.
- 3) Extensive experiments show that even with much smaller model sizes, SDConv achieves on-par or even better results compared with state-of-the-art methods for 3D shape generations both in terms of time complexity and reconstruction accuracy. The source code and pre-processed datasets have been formally released at: <https://github.com/Gaozhongpai/HSDConvMesh>.

This journal version extends and enhances our previous work [11] with significant developments and contributions as follows:

- 1) We introduce an adaptive sampling technique to refine the hierarchical levels of the template during training, substantially augmenting the model's representational capabilities (Section IV-B). On DFAUST, the error reduces from 3.822 mm to 3.400 mm ( $\downarrow 11.04\%$ ), as shown in Table V.
- 2) We further reform the input vertices feature of our hierarchical SDConv (HSDConv) to a balanced fusion of the center vertex and its neighbors (Section IV-A (5)). Our new design takes more consideration of the center vertex and thus improves the performance. On COMA, the error reduces from 0.159 mm to 0.154 mm ( $\downarrow 3.14\%$ ).
- 3) We extend our design to a mesh manipulation application in Section V-G (Fig. 11, Table VI, Figs. 12, and 13). By

doing simple manipulations such as adding, subtraction, or interpolation just on the feature, we can generate new meshes with interesting and explicit semantics. This further proves the representation ability of the learned latent feature.

- 4) We evaluate our new method more extensively, including more comparisons with recent methods, and a comparison on a new dataset of human hands.

## II. RELATED WORK

In this section, we discuss relevant literature across several key domains: linear 3D morphable models, graph neural networks, especially for 3D geometric data, and point neural networks. Moreover, we explore the related work on 3D shape correspondence and monocular 3D shape reconstruction. Finally, we discuss related work on dictionary embedding built into deep neural networks.

### A. Linear 3D Morphable Models

3D Morphable Models (3DMMs) serve as parametric frameworks for the representation of three-dimensional shapes such as human faces, bodies, and hands. These models are derived by conducting Principal Component Analysis (PCA) on a collection of 3D meshes, all of which adhere to a shared template. A notable application of 3DMM in facial modeling combines the Basel Face Model (BFM) [25], incorporating 200 individuals with neutral expressions, with the FaceWarehouse [6], featuring 150 individuals across 20 varied expressions. For body modeling, the Skinned Multi-Person Linear Model (SMPL) [26] stands out, offering a comprehensive representation of diverse body shapes in natural poses. Additionally, the MANO model [27] provides detailed modeling of human hands, based on roughly 1000 high-resolution 3D scans covering an extensive range of poses. These PCA-based constructs have been pivotal in reconstructing 3D faces, bodies, and hands. Our proposed SDConv method introduces the possibility of developing a nonlinear 3DMM, promising significantly enhanced representational capabilities for 3D shapes.

### B. Graph Neural Networks

In recent years, there's been a burgeoning interest in adapting deep learning techniques for graph-structured data. GCN [28] extends convolutional operations to graphs by employing fast localized and isotropic filters. GraphSage [29] innovates by sampling a fixed subset of neighbors to aggregate features for each vertex. GAT [30] introduces attention mechanisms to ascertain the relative importance between connected vertices. MoNet [31] utilizes pseudo-coordinates for neighboring vertices to apply varying weights, while FeaStNet [18] designs a graph convolutional operator that dynamically computes weighting matrices based on vertex features. These methodologies, though pioneering, do not specifically cater to mesh structures with a fixed topology and thus, may not directly compare with approaches that exploit this fixed topology advantage. LSA-Conv [12] emerged as a technique to learn weighting matrices

for each vertex within a template, facilitating the adaptation of CNNs to mesh data by resampling vertex neighbors. SIS [22] introduces an implicit continuous representation for mesh vertices, enabling smooth interpolation across the mesh surface; while this continuity is valuable for certain applications, it comes at the cost of reconstruction accuracy compared to discrete methods like LSA-small, the smaller version of LSA-Conv [12] on standard benchmarks. Our SDConv also focuses on constructing weighting matrices for vertices but leverages spectral dictionary learning principles to establish a distributed representation for each vertex, offering a nuanced approach to managing vertex features and their relational dynamics.

Recent work has also explored geometric regularization approaches for mesh generators. ARAPReg [32] regularizes the tangent space of shape manifolds using as-rigid-as-possible deformation models to improve generation quality. Building on similar deformation principles, GeoLatent [33] takes a differential geometry perspective to optimize the latent space of deformable shape generators, enforcing properties such as geodesic interpolation and pose-shape disentanglement through Riemannian metrics derived from ARAP (as-rigid-as-possible) and ACAP (as-conformal-as-possible) deformation energies. While these methods focus on regularization and latent space design rather than defining new convolution operations, they demonstrate the importance of geometric priors in mesh processing.

### C. Point Neural Networks

Point clouds represent a fundamental form of 3D geometric data. PointNet [34] introduced a groundbreaking architecture that processes each point individually through multi-layer perceptrons (MLPs), though it falls short in capturing the local structures inherent to point clouds. Subsequently, various methods for aggregating neighboring features [35], [36], [37] were developed to address this limitation. DGCNN [38] innovated with a convolutional operation focused on edge features to encapsulate relationships between points and their neighbors. PointCNN [23] introduced an  $\mathcal{X}$ -transformation learned from the input points to impose a form of order. RS-CNN [39] leveraged relation-shape convolutional operations, extracting high-level relational expressions from geometric priors. KP-Conv [40] applied convolutions over radius neighbors using weights positioned by kernel points, emphasizing spatial relations. Unlike FeaStNet [18], PointCNN [23], and KPConv [40], which dynamically compute weighting matrices based on input features, our SDConv method optimizes for the unique challenge posed by meshes with fixed topology. It does so by learning distributed representations for vertex weighting matrix bases, a strategy not applicable to the inherently unstructured nature of point cloud data.

### D. 3D Shape Correspondences

3D shape correspondence plays a pivotal role in numerous applications across computer vision and graphics, including shape interpolation, deformation, surface completion, and cross-shape texture mapping, as highlighted in [41] and [19]. This

task typically involves geometric 3D shapes represented either as triangle meshes or point clouds and encompasses processes often described as shape registration, alignment, or matching. Echoing the approach of [4], the objective of the 3D shape correspondence task in our study is to establish accurate correspondences between a 3D shape presented as a point cloud with randomized vertex order and a predefined common template shape. This is accomplished by encoding the point cloud into a low-dimensional global feature through a point cloud encoder and subsequently reconstructing the 3D shape from this global feature to align with the template's topology, facilitated by a mesh decoder.

### E. Monocular 3D Shape Reconstruction

Reconstructing 3D shapes from 2D imagery, particularly from a singular image, poses a complex challenge due to the ill-posed nature of the problem. Traditionally, 3DMMs serve as foundational priors to navigate this complexity. Yet, the reliance on linear 3DMMs constrains the diversity of achievable poses, expressions, and fails to account for occlusions, and lighting variations, among other factors. To enhance representational capacity, pioneering efforts by Tewari et al. [42], Tran et al. [43], and Guo et al. [44] have introduced dual-network 3D face models, consisting of a primary network for coarse adjustments via linear 3DMM and a secondary network for fine-grained corrections. Tran et al. [1], [45] progressed this concept by incorporating nonlinear activations and CNNs within the shape decoder to refine the face model further. Gao et al. [3] leveraged mesh convolutional operations within the shape decoder to elevate face shape generation capabilities. Similar mesh convolutional techniques have been applied to the domain of monocular 3D hand reconstruction [46]. Additionally, Line et al. [47] innovated by employing a Transformer [48] architecture for the shape decoder, facilitating the creation of 3D hands or bodies. Our proposed SDConv and HSDConv can be used for enhancing monocular 3D shape reconstruction tasks across faces, hands, and bodies, offering a versatile toolset for detailed and accurate model generation.

### F. Dictionary Learning

Dictionary learning is a method that expresses data as a linear combination of a limited set of basis elements from a dictionary matrix, with a coefficient vector specifying the linear combination's weights. This approach has been successfully integrated into deep neural networks in recent research. For instance, Liu et al. [49] innovated by substituting both the fully connected and activation layers with a dictionary learning layer in applications like scene recognition. Similarly, Mahdizadehaghdam et al. [50] developed a model that incorporates a hierarchical arrangement of dictionaries deep within the network structure for enhanced image classification.

Inspired by dictionary learning, this paper introduces a spectral dictionary to establish a distributed representation for each vertex within the convolutional layer, specifically tailored for mesh data. This spectral dictionary organizes the neighbors of

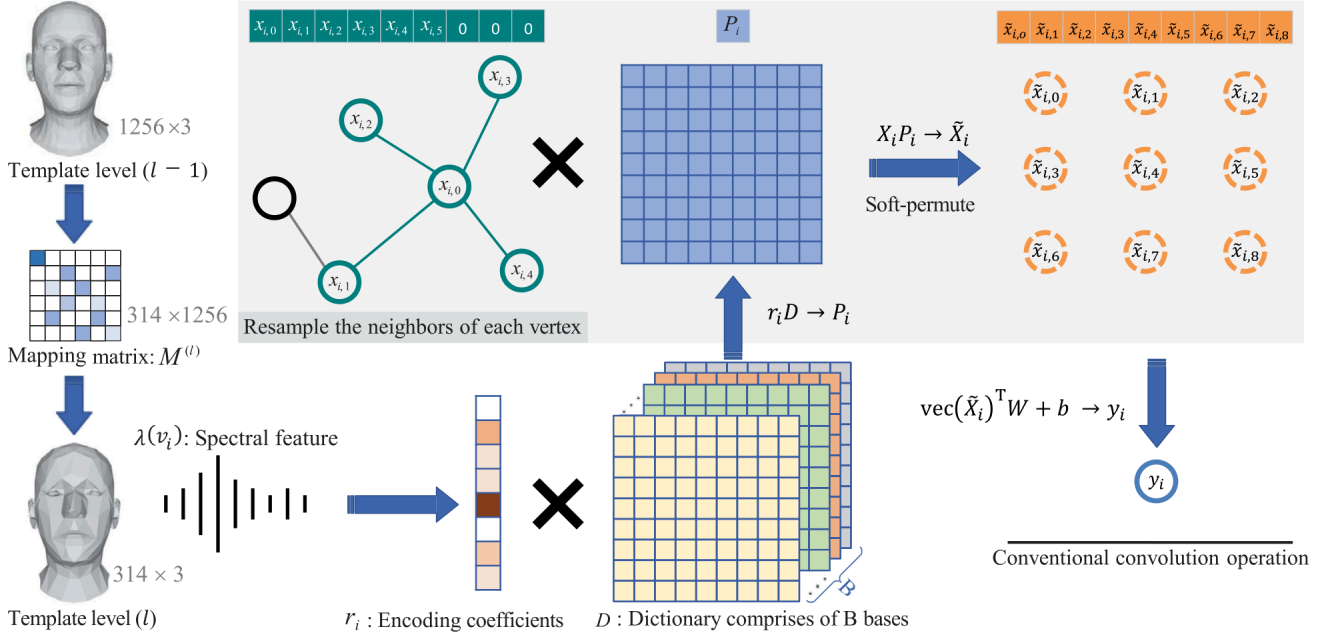


Fig. 3. The Hierarchical Spectral Dictionary Convolutional Operation (Hierarchical SDConv) extends the convolutional process to accommodate the unique structure of mesh data. For a given vertex  $x_i$  and its one-ring neighborhood  $\{x_{i,0}, x_{i,1}, \dots, x_{i,5}\} \subseteq \mathcal{N}_i$  (including the vertex itself), we organize  $X_i = \{x_{i,0}, x_{i,1}, \dots, x_{i,K-1}\} \in \mathbb{R}^{D_{in} \times K}$ , where  $K$  represents a predetermined number of neighbors. The vertex feature  $v_i \in \mathbb{R}^6$ , derived from template hierarchical levels, undergoes transformation via a spectral feature mapping function  $\lambda(v_i)$ . Our approach diverges from traditional methods by not learning individual weighting matrices for each vertex. Instead, we establish a set of  $B$  learnable bases  $D \in \mathbb{R}^{B \times K \times K}$  for the weighting matrices, alongside each vertex's corresponding coefficients  $r_i \in \mathbb{R}^B$ , derived from spectral features. The weighting matrix for each vertex is then assembled as  $P_i = r_i D$ , enabling the reordering of neighbors. Subsequently, conventional convolution is applied to the reorganized neighbors  $\tilde{X}_i = \{\tilde{x}_{i,0}, \tilde{x}_{i,1}, \dots, \tilde{x}_{i,9}\}$ , producing the output  $y_i$ . It's noteworthy that the template for each hierarchical level is generated by applying an adaptive mapping matrix  $M^{(l)}$  to the template of the preceding level, facilitating an iterative refinement process.

each vertex into a pseudo-canonical order, enabling the application of CNNs equipped with anisotropic filters to mesh structures. This approach not only bridges the gap between traditional CNN applications and 3D mesh processing but also leverages the geometric intricacies of meshes for enhanced representation and analysis.

### III. PRELIMINARIES

This section provides an overview of LSA-Conv [12] for context, maintaining consistency in mathematical notation for clarity. A 3D mesh is denoted as  $\mathcal{M} = (\mathcal{V}, \mathcal{E})$ , where  $\mathcal{V} = \{1, \dots, N\}$  represents the set of vertices with  $N$  indicating the total number of vertices, and  $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$  denotes the edges between these vertices. Each vertex is associated with attributes  $X \in \mathbb{R}^{D \times N}$ , with  $D$  being the dimensionality of the features (for instance,  $D = 3$  for coordinates in 3D space). The one-ring neighborhood of a vertex  $x_i$ , including the vertex itself as  $x_{i,0}$ , is represented by  $\mathcal{N}_i$ . This includes all neighbors  $x_{i,j} \in \mathcal{N}_i$  where  $(x_{i,0}, x_{i,j}) \in \mathcal{E}$ . We define  $\mathcal{N}_i = \{x_{i,0}, x_{i,1}, \dots, x_{i,|\mathcal{N}_i|-1}\}$ , with  $|\mathcal{N}_i|$  specifying the count of neighbors, which varies across the mesh's vertices.

For each vertex, we form  $X_i = \{x_{i,0}, x_{i,1}, \dots, x_{i,K-1}\} \in \mathbb{R}^{D_{in} \times K}$  by selecting the first  $K$  neighbors, or applying zero-padding if  $K > |\mathcal{N}_i|$ , as depicted in Fig. 3. Empirically,  $K$  is set

to 9 for all vertices, considering a practical balance between capturing sufficient neighbor information and ensuring computational efficiency, as validated in the experimental findings of [12] (refer to Fig. 6). This choice accommodates the typical range of 4-7 neighbors found in most mesh vertices. Moreover, standardizing the number of neighbors across all vertices to a uniform count simplifies the implementation for parallel computing on GPUs, enhancing the model's computational feasibility.

The sequence of neighbors  $\{x_{i,0}, x_{i,1}, \dots, x_{i,K-1}\}$  in  $X_i$  is initially arbitrary. To reorder the neighbors according to the vertex's local geometry, we use a learnable matrix  $P_i \in \mathbb{R}^{K \times K}$  (i.e., as model parameters), leading to the resampled neighborhood

$$\tilde{X}_i = X_i P_i, \quad (1)$$

where  $\tilde{X}_i \in \mathbb{R}^{D_{in} \times K}$ . This matrix  $P_i$  adapts to the specific geometric structure of each vertex's neighborhood. Subsequently, we apply a standard convolution operation:

$$y_i = \text{vec}(\tilde{X}_i)^T W + b, \quad (2)$$

with  $W \in \mathbb{R}^{(D_{in} \cdot K) \times D_{out}}$  representing  $D_{out}$  anisotropic filters,  $b \in \mathbb{R}^{D_{out}}$  as the bias term, and  $y_i \in \mathbb{R}^{D_{out}}$  as the output feature vector for the input vertex  $x_i \in \mathbb{R}^{D_{in}}$ . The operator  $\text{vec}(\cdot)$  flattens a matrix into a vector. Hence, LSA-Conv is encapsulated by

$$y_i = f(\text{vec}(f(X_i P_i))^T W + b), \quad (3)$$

TABLE I  
KEY NOTATION FOR SDConv AND HSDConv

| Symbol                                            | Meaning                                              |
|---------------------------------------------------|------------------------------------------------------|
| $N$                                               | Number of vertices in mesh                           |
| $K$                                               | Number of neighbors per vertex (fixed at 9)          |
| $B$                                               | Number of dictionary bases ( $B \ll N$ )             |
| $\mathbf{D} \in \mathbb{R}^{B \times K \times K}$ | Learnable weighting matrix bases (shared)            |
| $\mathbf{r}_i \in \mathbb{R}^B$                   | Coefficients for vertex $i$ (from spectral features) |
| $\mathbf{P}_i = \mathbf{r}_i \mathbf{D}$          | Weighting matrix for vertex $i$                      |
| $\mathbf{M}^{(l)}$                                | Mapping matrix at hierarchical level $l$             |
| $\mathbf{v}_i \in \mathbb{R}^6$                   | Vertex feature (center + neighbor edges)             |
| $\lambda(\mathbf{v}_i)$                           | Spectral feature mapping (preserves high-freq info)  |

employing  $f(\cdot)$ , an activation function such as ELU [51], to inject non-linearity into the model.

#### IV. APPROACH

*Overview:* Our method addresses two key limitations of LSA-Conv: (1) the model size grows linearly with vertex count since each vertex has its own weighting matrix, and (2) hierarchical sampling is static and predetermined. We address these through:

- *Spectral Dictionary (SDConv):* Instead of learning  $N$  separate weighting matrices, we learn  $B$  shared bases ( $B \ll N$ ) and vertex-specific coefficients derived from spectral features (Section IV-A).
- *Adaptive Sampling (HSDConv):* We learn hierarchical mapping matrices during training that can replace static preprocessing matrices at inference—adding no parameters (Section IV-B).

Table I summarizes key symbols used throughout this section.

##### A. Architecture Design for SDConv

*Motivation:* The LSA-Conv approach requires learning individual weighting matrices for each vertex, leading to substantial parameter counts that scale linearly with mesh resolution. For example, with  $N = 5023$  vertices and  $K = 9$  neighbors, the total parameter count reaches  $N \times K \times K = 406,863$ . This becomes prohibitive for high-resolution meshes.

*Key Idea:* Given the geometric similarity across many vertices, learning a unique weighting matrix per vertex is inefficient. Instead, we posit that weighting matrices reside within a confined subspace and can be reconstructed through linear combinations of a small set of basis matrices—analogueous to dictionary learning. This reduces parameters from  $O(N)$  to  $O(1)$  with respect to mesh resolution.

*Spectral Dictionary Decomposition:* For mesh with  $N$  vertices, we decompose all weighting matrices  $\mathbf{P} \in \mathbb{R}^{N \times K \times K}$  as:

$$\mathbf{P} = \mathbf{R}\mathbf{D}, \quad (4)$$

where:

- $\mathbf{D} \in \mathbb{R}^{B \times K \times K}$  is the *spectral dictionary* composed of  $B$  learnable basis matrices, shared across all vertices ( $B \ll N$ , e.g.,  $B = 8$  for  $N = 5023$ )
- $\mathbf{R} = \{\mathbf{r}_0, \mathbf{r}_1, \dots, \mathbf{r}_{N-1}\} \in \mathbb{R}^{N \times B}$  contains vertex-specific coefficients

Crucially,  $\mathbf{D}$  comprises shared model parameters (only  $B \times K \times K$  values), while coefficients  $\mathbf{R}$  are *computed* from each vertex's spectral features rather than stored as parameters. This distributed representation dramatically reduces model size.

*Computing Coefficients from Spectral Features:* The coefficients  $\mathbf{r}_i \in \mathbb{R}^B$  for each vertex are derived from the geometric structure of the template and its hierarchical levels (Fig. 3). The template's hierarchical levels are generated using our adaptive mapping matrices (Section IV-B). We compute the feature vector for each template vertex as:

$$\mathbf{v}_i = \left\{ \mathbf{t}_{i,0} \oplus \frac{\sum_{j=1}^{|\mathcal{N}_i|-1} (\mathbf{t}_{i,0} - \mathbf{t}_{i,j})}{|\mathcal{N}_i| - 1} \right\}, \quad (5)$$

where  $\mathbf{v}_i \in \mathbb{R}^6$  represents a balanced fusion of the center vertex position ( $\mathbf{t}_{i,0} \in \mathbb{R}^3$ ) and the mean edge displacement to its neighbors ( $\in \mathbb{R}^3$ ). Here,  $\mathbf{t}_{i,j} \in \mathbb{R}^3$  is the  $j$ th neighbor of vertex  $i$  in the template, and  $\oplus$  denotes concatenation. This balanced representation equally weights the vertex's absolute position and its local geometric context.

To preserve high-frequency geometric details, we apply a spectral feature mapping that projects  $\mathbf{v}_i$  onto a higher-dimensional feature space using sinusoidal functions:

$$\lambda(\mathbf{v}_i) = \{\cos(2\pi \mathbf{v}_i \mathbf{G}) \oplus \sin(2\pi \mathbf{v}_i \mathbf{G})\}, \quad (6)$$

where  $\mathbf{G} \in \mathbb{R}^{6 \times M}$  is a fixed (non-trainable) Gaussian random matrix, and  $M$  is the spectral feature dimension. This random Fourier feature mapping [52] enables the network to capture high-frequency variations in local geometry.

Finally, the basis coefficients are computed by encoding the spectral features:

$$\mathbf{r}_i = g(\text{Linear}_1(\lambda(\mathbf{v}_i))), \quad (7)$$

where  $g(\cdot)$  is a softmax function with adaptive temperature (detailed in Section IV-C), and  $\text{Linear}_1(\cdot)$  is a linear layer with weights  $\mathbf{W}_1 \in \mathbb{R}^{B \times M}$ .

*Parameter Count Comparison:* This approach diverges fundamentally from LSA-Conv, which directly learns coefficients  $\mathbf{r}_i$  as model parameters. Instead, we learn only the shared weights  $\mathbf{W}_1$  that map spectral features to coefficients—applicable across the entire template. The total parameter count for weighting matrices becomes:

$$\text{SDConv: } B \times K \times K + M \times B = 1,160$$

$$(B = 8, K = 9, M = 32)$$

$$\text{LSA-Conv: } N \times K \times K = 406,863 \quad (N = 5023, K = 9)$$

representing a **350× reduction** in parameters. The formulation for SDConv is:

$$\mathbf{y}_i = f(\text{vec}(f(\mathbf{X}_i \cdot \mathbf{r}_i \cdot \mathbf{D})))^\top \mathbf{W} + \mathbf{b}), \quad (8)$$

where  $\mathbf{W} \in \mathbb{R}^{(D_{in} \cdot K) \times D_{out}}$  represents the convolutional filters,  $\mathbf{b} \in \mathbb{R}^{D_{out}}$  is the bias, and  $f(\cdot)$  is the activation function (e.g., ELU).

*Skip Connection:* To enhance gradient flow and emphasize the importance of each vertex's own features, we incorporate a residual connection:

$$\mathbf{y}_i = f(\text{vec}(f(\mathbf{X}_i \cdot \mathbf{r}_i \cdot \mathbf{D})))^\top \mathbf{W} + \mathbf{b}) + \text{Linear}_2(\mathbf{x}_i), \quad (9)$$

where  $\text{Linear}_2(\mathbf{x}_i)$  employs  $\mathbf{W}_2 \in \mathbb{R}^{D_{in} \times D_{out}}$  to align input and output channel dimensions.

### B. Adaptive Hierarchical Mapping Matrices

*Motivation:* Hierarchical mesh processing requires sampling operations to aggregate features across different resolutions. Prior work [11], [12], [13] uses predetermined mapping matrices  $\mathbf{M}_0^{(l)}$  computed via preprocessing (e.g., quadric error metrics), which are static and cannot adapt during training. This limits the network’s ability to learn optimal feature aggregation strategies for the task at hand.

*Key Idea:* We propose learning the hierarchical mapping matrices during training while maintaining *zero additional parameters at inference*. This is achieved by blending a fixed preprocessing matrix with a learned correction during training, then using only the final blended matrix at inference.

*Hierarchical Feature Aggregation:* The encoder processes a 3D mesh through progressively coarser levels to extract a compact latent feature, while the decoder reconstructs mesh details through gradual refinement. At each hierarchical level  $l \in \{1, 2, \dots, L\}$ , features  $\mathbf{X}^{(l)} \in \mathbb{R}^{D_l \times N_l}$  are obtained by applying a mapping matrix to features from the previous level  $\mathbf{X}^{(l-1)} \in \mathbb{R}^{D_{l-1} \times N_{l-1}}$ :

$$\mathbf{X}^{(l)} = \mathbf{X}^{(l-1)} \cdot \mathbf{M}^{(l)\top}, \quad (10)$$

where  $\mathbf{M}^{(l)} \in \mathbb{R}^{N_l \times N_{l-1}}$  aggregates  $N_{l-1}$  vertices into  $N_l$  vertices ( $N_l < N_{l-1}$  for downsampling,  $N_l > N_{l-1}$  for upsampling).

*Training Phase: Learning Adaptive Mappings:* During training, we construct  $\mathbf{M}^{(l)}$  as a learned blend of a static preprocessing matrix and a trainable blend:

$$\mathbf{M}^{(l)} = \frac{w_0 \cdot \mathbf{M}_0^{(l)} + w_1 \cdot \text{top}k(\tanh(\mathbf{M}_1^{(l)}))}{w_0 + w_1}, \quad (11)$$

where:

- $\mathbf{M}_0^{(l)} \in \mathbb{R}^{N_l \times N_{l-1}}$  is the *fixed* preprocessing matrix from [13]
- $\mathbf{M}_1^{(l)} \in \mathbb{R}^{N_l \times N_{l-1}}$  is *learnable* during training only
- $w_0, w_1 \in \mathbb{R}$  are learnable scalar blend weights
- $\text{top}k(\cdot)$  retains the top  $k$  values per row, zeros others, then normalizes—preserving sparsity like  $\mathbf{M}_0^{(l)}$

The  $\tanh$  nonlinearity bounds  $\mathbf{M}_1^{(l)}$  values, while  $\text{top}k$  ensures the learned matrix maintains local connectivity similar to the preprocessing matrix. The blend weights  $w_0$  and  $w_1$  control the balance between the fixed geometric prior and learned adaptation.

*Inference Phase: Zero-Parameter Drop-In Replacement:* Once training converges, the mapping matrix  $\mathbf{M}^{(l)}$  is *fixed* and directly replaces  $\mathbf{M}_0^{(l)}$  in (10). Critically, the training parameters  $\mathbf{M}_1^{(l)}$ ,  $w_0$ , and  $w_1$  are discarded after training as they served to learn  $\mathbf{M}^{(l)}$  but contribute zero parameters at inference. Since  $\mathbf{M}^{(l)}$  maintains the same sparsity pattern as  $\mathbf{M}_0^{(l)}$ , it can be stored and applied identically, requiring no architectural changes or additional memory.

*Dynamic Spectral Features:* The adaptive templates  $\mathcal{T}^{(l)}$  generated by (10) provide the vertex features  $\mathbf{v}_i$  (5) that determine

SDConv’s basis coefficients  $\mathbf{r}_i$  (7). By refining hierarchical templates during training, the spectral features adapt to better capture geometric variation across scales. This is particularly beneficial for datasets with diverse poses (e.g., DFAUST bodies) where predetermined sampling may miss task-relevant structure. The synergy between adaptive sampling and spectral dictionary learning yields our hierarchical SDConv (**HSDConv**), which simultaneously optimizes both the dictionary bases and the hierarchical feature extraction pathway.

### C. Adaptive Temperature for Softmax

*Motivation:* In (7), the basis coefficients  $\mathbf{r}_i$  are computed via softmax, which determines how each vertex combines the  $B$  dictionary bases. Standard softmax produces relatively uniform distributions, but different geometric regions may benefit from sparser or softer combinations of bases.

*Key Idea:* We introduce an adaptive temperature  $T_i$  learned from each vertex’s spectral features, allowing the network to control the “sharpness” of basis selection based on local geometry.

The coefficients are computed as:

$$\mathbf{r}_{i,j} = g(\mathbf{z}_{i,j}) = \frac{\exp(\mathbf{z}_{i,j}/T_i)}{\sum_{k=0}^{B-1} \exp(\mathbf{z}_{i,k}/T_i)}, \quad (12)$$

where  $\mathbf{z}_i = \text{Linear}_1(\lambda(\mathbf{v}_i))$  and the temperature is computed as:

$$T_i = s(\text{Linear}_3(\lambda(\mathbf{v}_i))) \times (0.1 - 1.0/F) + 1.0/F, \quad (13)$$

with  $s(\cdot)$  being the sigmoid activation and  $F = 100$ , ensuring  $T_i \in [0.01, 0.1]$ . The layer  $\text{Linear}_3(\cdot)$  uses weights  $\mathbf{W}_3 \in \mathbb{R}^{1 \times M}$ .

Since  $T_i < 1$ , the resulting distributions are *sharper* than standard softmax (lower entropy), producing more selective basis combinations. This encourages sparse, interpretable representations where each vertex primarily uses a small subset of the dictionary bases, analogous to sparse coding in classical dictionary learning.

## V. EXPERIMENTS AND EVALUATION

Initially, we assess our method across two distinct tasks on different 3D shape datasets: autoencoder-based reconstruction and 3D shape correspondence. In the autoencoder-based reconstruction task, the input consists of 3D meshes with a fixed topology, whereas, for the 3D shape correspondence task, we employ point clouds with a randomized arrangement of vertices. Subsequently, we broaden our evaluation to monocular 3D hand reconstruction, illustrating a practical deployment of our technique. Further, through ablation studies, we substantiate the effectiveness of both SDConv and HSDConv. Additionally, we provide visualizations of the learned hierarchical mapping matrices and weighting matrices to offer insights into the model’s internal workings. Finally, we showcase the capability of our learned latent feature in various manipulation tasks, demonstrating the robust representation power of our approach.

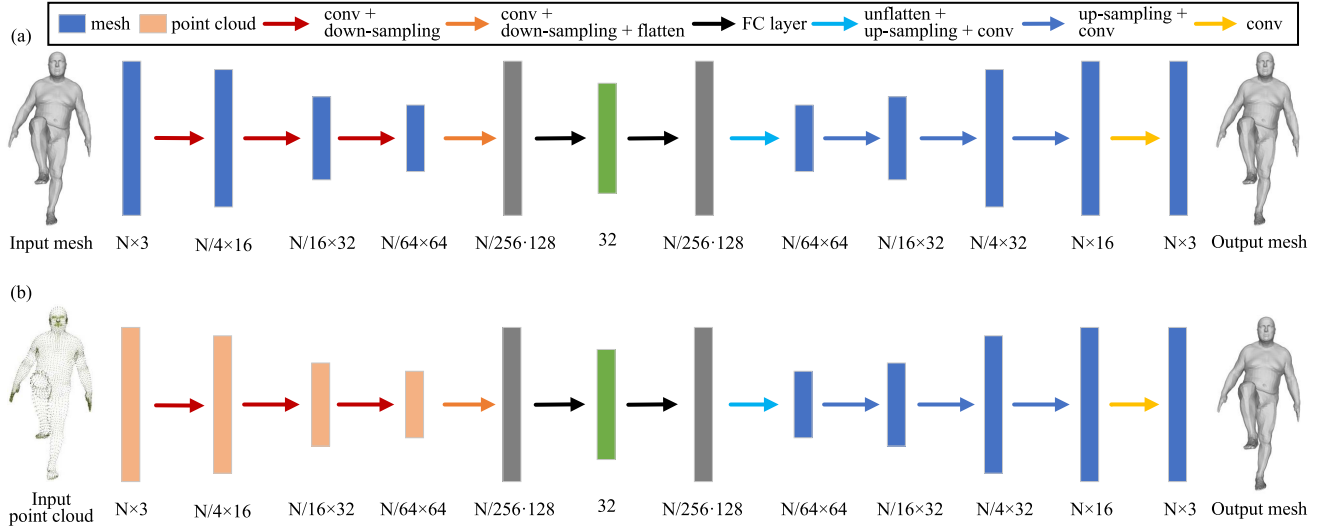


Fig. 4. Overview of the autoencoder for reconstruction and 3D shape correspondence tasks. The architecture facilitates processing of meshes with  $N$  vertices, through both downsampling and upsampling, utilizing either predefined [13] or the introduced adaptive mapping matrices. (a) In the reconstruction task, convolutional layers may employ techniques such as COMA, Spiral, Spiral++, Adaptive-Spiral, VCMeshConv, LSA-Conv, FeaStNet, or SDConv for comparisons. (b) For the 3D shape correspondence task, the encoder's convolutional layers are substituted with PAI-Conv [53], a specialized convolution operation designed for point cloud data. Downsampling within the encoder relies on a random sampling approach to accommodate point clouds effectively.

#### A. Protocol

**Datasets:** For evaluation on autoencoder-based reconstruction and 3D shape correspondence tasks, we use two datasets as per [12]: COMA [13] and DFAUST [19]. COMA, a human facial dataset, comprises 12 extreme expression classes from 12 distinct subjects, totaling 20,466 3D meshes aligned to a standard template of 5,023 vertices. DFAUST, on the other hand, is a human body dataset featuring over 40,000 real meshes that document 129 dynamic performances across 10 subjects, each mesh conformed to a reference topology of 6,890 vertices. We partition both datasets into training and testing sets with a 9:1 ratio and select a random subset of 100 samples from the training data for validation purposes. To facilitate efficient training, we align the poses of ground-truth meshes with the templates and calculate the mean and standard deviation of the 3D meshes over the training set to normalize the 3D meshes.

For the task of 3D hand reconstruction, we assess our approach using the FreiHAND dataset [24]. FreiHAND encompasses a collection of 3D hand images, with the training set containing 130,000 images and the test set comprising 4,000 images. Evaluations are carried out via the dataset's online server.

**Training:** For both the autoencoder-based reconstruction and 3D shape correspondence tasks, the Adam optimizer [54] is employed, starting with a learning rate of 0.001 and applying a decay rate of 0.99 to the learning rate at the end of each epoch. We set the batch size to 32 and the total of epochs to 300. The network parameters are subject to weight decay regularization, except the spectral dictionary associated with the weighting matrices. In alignment with the methodologies presented in [12], the bases of the weighting matrices are initialized to identity matrices. The training regimen for all experiments relies solely on the L1 reconstruction loss and the models' performance is quantitatively assessed using L2 errors measured in millimeters (mm).

For the task of 3D hand reconstruction, the Adam optimizer [54] is utilized, set at a learning rate of 0.0001. We set a batch size of 192 and the total of epochs to 50. Training across all experiments incorporates L1 losses for 3D vertices, 3D joints, and 2D projected joints. Evaluation metrics include the Mean-Per-Joint-Position-Error [55] and Mean-Per-Vertex-Position-Error, both employing Procrustes Analysis [56] (PAMPJPE and PA-MPVPE, respectively), alongside the F-score at 5 mm (F@5 mm) and F-score at 15 mm (F@15 mm) to gauge the accuracy of the reconstructions.

We conducted comparisons across various convolutional operations, all implemented in PyTorch and executed on identical hardware comprising an AMD 3700X @3.6 GHz CPU and an NVIDIA RTX2080Ti GPU.

**Architecture:** For the autoencoder-based reconstruction task, we utilize the network architecture outlined in [12], with the convolutional layer options including our new SDConv, SDConv\* [11], LSA-Conv and LSA-small [12], COMA [13], Spiral [14], SpiralNet++ [15], Adaptive-Spiral [16], VCMeshConv [17], FeaStNet [18], and AGConv [57] for the respective methodologies. This network follows an encoder-decoder structure, featuring a latent vector centrally positioned, as depicted in Fig. 4. Initially, the mesh encoder comprises four convolutional layers accompanied by downsampling, with the layers' channel sizes set at [3, 16, 32, 64, 128] and downsampling ratios applied at [4, 4, 4, 4]. Subsequently, a fully connected layer generates the latent feature representing the 3D mesh. The decoder part consists of five convolutional layers designed for upsampling, where the channel sizes are arranged as [128, 64, 32, 32, 16, 3] and the upsampling ratios are [4, 4, 4, 4, 1]. Ultimately, the decoder reconstructs and outputs the 3D mesh. Both down- and upsampling processes leverage either the predefined mapping matrices [13] or the novel adaptive mapping matrices introduced in our methodology.

In the task of 3D shape correspondence, we modify the encoder's convolutional layer within the architecture shown in Fig. 4 to utilize PAI-Conv [53], a convolutional operation optimized for point cloud data. To achieve the downsampling of point clouds, we employ a random sampling technique, maintaining the same downsampling ratios of [4, 4, 4, 4] as in the autoencoder reconstruction task.

In the 3D hand reconstruction task, we employ ResNet-18 [58] as the backbone to derive a global feature from the input image. This global feature then serves as the input for a fully connected layer, which generates a latent feature of dimension  $d = 128$ , encapsulating the hand's characteristics. Additionally, an MLP with channel dimensions of [128, 7] is utilized to determine the quaternion-based camera parameters. For the reconstruction of the 3D hand mesh, a dedicated mesh decoder equipped with four convolutional layers and three stages of upsampling processes the latent feature. These convolutional layers feature channel dimensions of [128, 64, 32, 32, 3] and kernel sizes of [6, 6, 6, 6]. The upsampling within the mesh reconstruction follows ratios of [4, 4, 4, 1], facilitating detailed mesh recovery.

### B. Autoencoder-Based Reconstruction

In the reconstruction task, we benchmark our approaches, SDConv and HSDConv, against established methods: LSA-Conv and its variant LSA-small [12], COMA [13], Spiral [14], SpiralNet++ [15], Adaptive-Spiral [16], VCMeshConv [17], FeaStNet [18], and AGConv [57]. These comparisons span various dimensions of the latent feature ( $d = 8, 16, 32, 64, 128$ ) while maintaining a consistent architecture. Notably, LSA-small is a parameter-reduction version of LSA-Conv. As illustrated in Fig. 5, our SDConv method demonstrates superior performance in minimizing reconstruction errors across both the COMA and DFAUST datasets for all tested latent feature dimensions. While SDConv exhibits slightly higher reconstruction errors than LSA-Conv, it compensates with a significantly reduced model size. Detailed comparisons with additional methods for the latent feature size of  $d = 32$ , are further elaborated in Table II.

To enhance SDConv, we introduce the hierarchical SDConv (HSDConv), which incorporates adaptive hierarchical mapping matrices for both downsampling and upsampling processes. Demonstrated in Fig. 5, HSDConv achieves superior performance compared to SDConv\* [11] across various latent feature dimensions on both the DFAUST and COMA datasets. Remarkably, on DFAUST, HSDConv not only outperforms SDConv but also achieves lower reconstruction errors than LSA-Conv, while maintaining a significantly reduced model size. The comparative analysis, as seen in Fig. 5 and Table II, indicates a more pronounced improvement of HSDConv over SDConv on the DFAUST dataset than on the COMA dataset. This difference is attributed to the greater diversity in human body poses compared to facial expressions, highlighting the advantage of employing adaptive hierarchical mapping matrices for handling varied body poses.

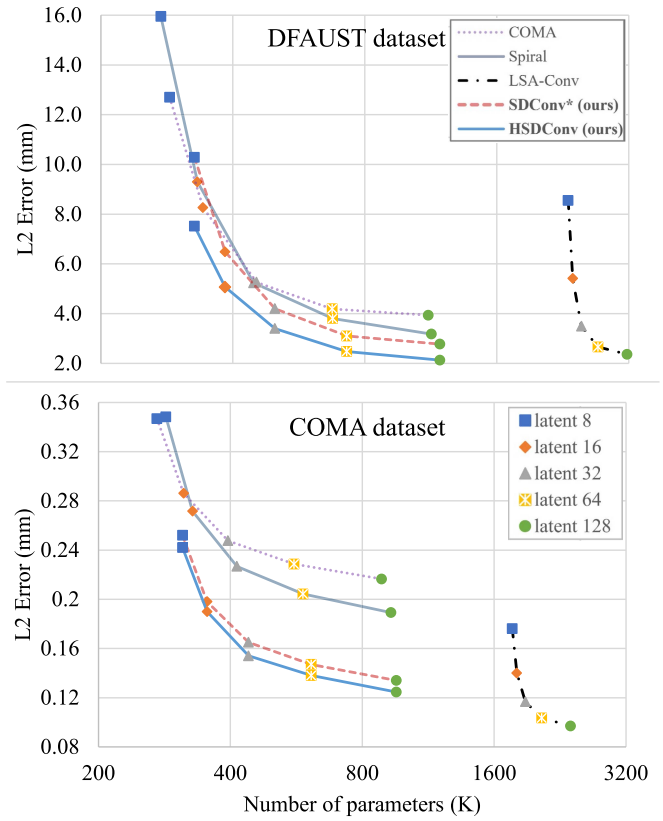


Fig. 5. Evaluation of SDConv\* [11] and HSDConv against peer methods: COMA [13], Spiral [14], and LSA-Conv [12] with different dimensionalities of the latent feature on the test sets of DFAUST and COMA dataset.

For a fair comparison, we calibrated the channel dimensions across all methods to achieve comparable model sizes, as detailed in Table II. Due to its inherently large model size, LSA-Conv was excluded from this adjustment. In this standardized setting, both SDConv and HSDConv outperform the established methods across the DFAUST and COMA datasets. As depicted in Fig. 1, concerning time complexity, our SDConv and HSDConv match the performance of LSA-Conv and Adaptive-Spiral, and surpass the efficiency of both COMA and FeaStNet. Furthermore, in Fig. 6(a), a qualitative analysis of reconstruction errors from five test set examples, with the latent dimension set to  $d = 32$ , reveals that HSDConv delivers superior visual quality in comparison to the competing methods.

Additionally, we explore an alternative strategy to achieve comparable model sizes for our HSDConv by adjusting the base size of the weighting matrices, as detailed in Table II. This adjustment also allows HSDConv to surpass traditional methods in performance across both DFAUST and COMA datasets. Notably, this approach of modifying the base sizes results in a marginally improved performance on the COMA dataset and a slight decrease on the DFAUST dataset, compared to adjustments in channel sizes. Thus, the choice between altering channel sizes or base sizes can be tailored to suit different mesh shape characteristics, allowing for flexibility in optimizing model performance and efficiency based on the specific requirements of the task at hand.



Fig. 6. Qualitative evaluation of our SDCov and HSDConv against: COMA [13], Spiral [14], and LSA-small [12] on DFAUST and COMA dataset when the latent size is  $d = 32$ . (a) Reconstruction task with around the same number of parameters (see quantitative results in Table II) and (b) 3D correspondence task (see quantitative results in Table III).

TABLE II

COMPARISON OF RECONSTRUCTION ERRORS FROM PCA [59], FEASTNET [18], AGCONV [57], LSA-CONV [12], COMA [13], SPIRAL [14], SPIRALNET++ [15], ADAPTIVE-SPIRAL [16], VCMESHCONV [17], AND OUR SDCONV AND HSDCONV WHEN LATENT SIZE  $d = 32$  AND DICTIONARY BASE SIZE  $B = 32$ . SDCONV\* DENOTES THE VERSION FROM OUR CONFERENCE PAPER [11].

LSA-SMALL IS A PARAMETER-REDUCTION VERSION OF LSA-CONV. VCMESHCONV [17] DOES NOT SUPPORT THE COMA DATASET BECAUSE THE FACE MESH HAS MORE THAN ONE COMPONENT: A HEAD AND TWO EYES. FEASTNET [18] AND AGCONV [57] SUPPORT BOTH FIXED OR ARBITRARY TOPOLOGIES. FOR A FAIR COMPARISON, WE INCREASE THE CHANNEL SIZES OR BASE SIZES TO HAVE AROUND THE SAME MODEL SIZE, AS LISTED ABOVE.

|                       | DFAUST                                            |               | COMA         |        |
|-----------------------|---------------------------------------------------|---------------|--------------|--------|
|                       | L2(mm)                                            | parm #        | L2(mm)       | parm # |
| PCA                   | 9.977                                             | 661K          | 0.210        | 482K   |
|                       | same architecture                                 |               |              |        |
| FeaStNet              | 6.028                                             | 441K          | 0.293        | 382K   |
| AGConv                | 5.537                                             | 472K          | 0.348        | 414K   |
| COMA                  | 5.238                                             | 361K          | 0.248        | 303K   |
| Spiral                | 5.258                                             | 446K          | 0.227        | 414K   |
| Spiral++              | 5.491                                             | 437K          | 0.212        | 379K   |
| Adaptive-Spiral       | 4.289                                             | 477K          | 0.186        | 416K   |
| VCMeshConv            | 4.162                                             | 2,646K        | N/A          |        |
| LSA-Conv              | 3.492                                             | 2,478K        | <b>0.117</b> | 1,867K |
| LSA-small             | 4.569                                             | 1,263K        | 0.176        | 986K   |
| <b>SDConv* (ours)</b> | 3.884                                             | 518K          | 0.159        | 459K   |
| <b>HSDConv (ours)</b> | <b>3.400</b>                                      | 518K          | 0.156        | 459K   |
|                       | ↑ channel size for around same # of param         |               |              |        |
| LSA-small (B=8)       | 4.544                                             | 644K          | 0.179        | 532K   |
| COMA                  | 5.110                                             | 658K          | 0.198        | 532K   |
| Spiral                | 4.667                                             | 647K          | 0.193        | 533K   |
| Spiral++              | 4.836                                             | 643K          | 0.169        | 533K   |
| Adaptive-Spiral       | 3.826                                             | 643K          | 0.157        | 534K   |
| VCMeshConv (B=4)      | 6.274                                             | 650K          | N/A          |        |
| <b>SDConv* (ours)</b> | 3.673                                             | 643K          | 0.148        | 531K   |
| <b>HSDConv (ours)</b> | <b>3.254</b>                                      | 643K          | 0.145        | 531K   |
|                       | ↑ base size for around same # of param            |               |              |        |
|                       | B=112                                             |               | B=78         |        |
| <b>HSDConv (ours)</b> | 3.372                                             | 643K          | <b>0.142</b> | 530K   |
|                       | channel sizes for around same # of param          |               |              |        |
| LSA-small (B=8)       | [3, 16, 32, 64, 128] - [128, 64, 32, 16, 3]       |               |              |        |
|                       | encoder                                           | <b>DFAUST</b> | decoder      |        |
| COMA                  | [3, 64, 96, 112, 128] - [128, 112, 96, 96, 64, 3] |               |              |        |
| Spiral                | [3, 32, 64, 64, 128] - [128, 110, 64, 64, 32, 3]  |               |              |        |
| Spiral++              | [3, 32, 64, 80, 128] - [128, 112, 64, 64, 36, 3]  |               |              |        |
| Adaptive-Spiral       | [3, 32, 64, 64, 128] - [128, 96, 68, 64, 36, 3]   |               |              |        |
| <b>HSDConv (ours)</b> | [3, 32, 45, 80, 128] - [128, 80, 64, 45, 32, 3]   |               |              |        |
|                       | encoder                                           | <b>COMA</b>   | decoder      |        |
| COMA                  | [3, 50, 64, 112, 128] - [128, 112, 96, 64, 50, 3] |               |              |        |
| Spiral                | [3, 32, 48, 64, 128] - [128, 96, 64, 48, 32, 3]   |               |              |        |
| Spiral++              | [3, 48, 60, 64, 128] - [128, 96, 64, 64, 32, 3]   |               |              |        |
| Adaptive-Spiral       | [3, 48, 60, 64, 128] - [128, 96, 64, 48, 32, 3]   |               |              |        |
| <b>HSDConv (ours)</b> | [3, 32, 45, 64, 128] - [128, 80, 48, 42, 32, 3]   |               |              |        |

### C. 3D Shape Correspondence

For the 3D shape correspondence task, we utilize PAI-Conv [53], an efficient convolutional operation tailored for point clouds, to extract a global feature from 3D shapes with randomized vertex order. The latent feature dimension is set at  $d = 32$ . To evaluate the representational capabilities of various convolutional operations, we reconstruct the 3D shape from its global feature using a mesh decoder configured with each convolutional operation, maintaining a consistent architecture as depicted in Fig. 4(b). This ensures the reconstructed shape retains the same topology as the template, enabling the identification of

TABLE III

COMPARISON OF 3D CORRESPONDENCE ERRORS FOR MODELS OF COMA [13], SPIRAL [14], LSA-SMALL [12], AND OUR SDCONV\* [11] AND HSDCONV WHEN LATENT SIZE  $d = 32$

|                       | DFAUST        |        | COMA         |        |
|-----------------------|---------------|--------|--------------|--------|
|                       | L2(mm)        | parm # | L2(mm)       | parm # |
| COMA                  | 20.564        | 475K   | 0.341        | 417K   |
| Spiral                | 21.398        | 475K   | 0.341        | 417K   |
| LSA-small             | 17.085        | 607K   | 0.312        | 514K   |
| <b>SDConv* (ours)</b> | 16.478        | 521K   | 0.294        | 463K   |
| <b>HSDConv (ours)</b> | <b>15.190</b> | 525K   | <b>0.285</b> | 465K   |

TABLE IV

COMPARISON OF MONOCULAR 3D HAND RECONSTRUCTION FOR THE MODELS OF COMA [13], SPIRAL [14], LSA-SMALL [12], AND OUR HSDCONV, EVALUATED ON FREIHAND ONLINE SERVER. WE TRAIN THE MODELS WITH SMALL BACKBONE (RESNET-18) AND DECODERS FOR EFFICIENCY.

|                       | PA-MPJPE↓   | PA-MPVPE↓   | F@5 mm↑      | F@15 mm↑     |
|-----------------------|-------------|-------------|--------------|--------------|
| COMA [13]             | 12.3        | 13.5        | 0.454        | 0.937        |
| Spiral [14]           | 11.1        | 10.6        | 0.540        | 0.947        |
| LSA-small [12]        | 11.4        | 10.9        | 0.532        | 0.945        |
| <b>HSDConv (ours)</b> | <b>10.5</b> | <b>10.2</b> | <b>0.559</b> | <b>0.952</b> |

correspondences between two shapes. According to the results listed in Table III, both HSDConv and SDConv outperform LSA-small, Spiral, and COMA in terms of accuracy. Echoing the findings from the autoencoder-based reconstruction task, HSDConv emerges as the top performer, showcasing superior representation capabilities for 3D shape generation. Qualitative comparisons of reconstruction errors from five test set examples, illustrated in Fig. 6(b), further affirm HSDConv's exceptional visual quality relative to competing methods. Notably, decoders powered by HSDConv and SDConv hold potential for broader applications in 3D shape reconstruction from images and other related 3D shape generation tasks.

### D. Monocular 3D Hand Reconstruction

In this section, we assess our approach within the context of monocular 3D hand reconstruction on the FreiHAND dataset [24]. As indicated in Table IV, HSDConv surpasses COMA, Spiral, and LSA-small across all evaluation metrics, showcasing its efficacy. The qualitative results, depicted in Fig. 7, further demonstrate our method's superiority in accurately reconstructing key points and the overall mesh. Notably, meshes reconstructed by the COMA method exhibit significant noise, resulting in lower visual quality, in contrast to the cleaner and more accurate reconstructions achieved by our HSDConv method.

### E. Ablation Study

The implementation of our adaptive temperature softmax, as detailed in (13), plays a crucial role in determining the coefficients for the weighting matrix dictionary, or bases. Table V illustrates that the utilization of the standard softmax function (index A) significantly increases the reconstruction errors across both datasets. Conversely, employing sparsemax (index B) yields lower reconstruction errors compared to the vanilla

TABLE V

ABLATION TESTS OF OUR METHODS WITH LATENT SIZE  $d = 32$  AND BASE SIZE  $B = 32$  ON DFAUST AND COMA. ROW “C - SDConv\*” IS OUR CONFERENCE VERSION RESULT [11], AND ROW “G - SDConv” AND “H - HSDConv” ARE OUR PROPOSED METHOD’S RESULTS IN THIS PAPER.

| index       | softmax design |        |          | spectral feature |                    | skip connection | adaptive hierarchical | DFAUST L2(mm) | COMA L2(mm)  |
|-------------|----------------|--------|----------|------------------|--------------------|-----------------|-----------------------|---------------|--------------|
|             | vanilla        | sp-max | adaptive | $v_i$            | balanced $G$ fixed |                 |                       |               |              |
| A           | ✓              |        |          | ✗                | ✓                  | ✓               |                       | 4.611         | 0.175        |
| B           |                | ✓      |          | ✗                | ✓                  | ✓               |                       | 4.157         | 0.164        |
| C - SDConv* |                |        | ✓        | ✗                | ✓                  | ✓               |                       | 3.884         | 0.159        |
| D           |                |        | ✓        | ✗                | ✓                  |                 |                       | 4.202         | 0.165        |
| E           |                |        | ✓        |                  | w/o                | ✓               |                       | 4.387         | 0.183        |
| F           |                |        | ✓        | ✓                | ✗                  | ✓               |                       | 3.850         | 0.154        |
| G - SDConv  |                |        | ✓        | ✓                | ✓                  | ✓               |                       | 3.832         | <b>0.154</b> |
| H - HSDConv |                |        | ✓        | ✓                | ✓                  | ✓               | ✓                     | <b>3.400</b>  | 0.156        |

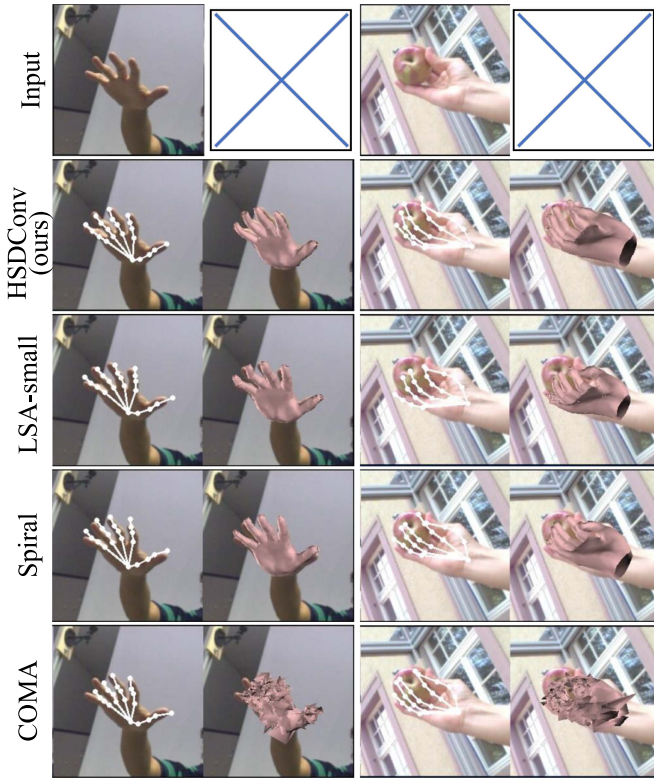


Fig. 7. Qualitative results of our method against COMA [13], Spiral [14], and LSA-small [12] on FreiHAND dataset.

softmax but fails to achieve the precision of the complete model, which incorporates the adaptive temperature softmax (index C). The adaptability of the softmax temperature, derived from the spectral features of each vertex and its adjacent neighbors, tailors the ‘softness’ of the coefficients’ output distribution for the weighting matrix bases, optimizing performance.

To learn the encoding coefficients, we employ a mapping from the vertex feature to the spectral feature as outlined in (6), a strategy instrumental in retaining high-frequency information. Omitting the spectral feature mapping function (index E) results in increased reconstruction errors on both datasets. It indicates that the spectral feature mapping technique significantly enhances the regression accuracy for the coefficients associated

with the weighting matrix bases, underscoring its importance in our approach.

In our exploration of vertex feature  $v_i$  and the Gaussian random mapping matrix  $G$  designs, we refine the approach taken in SDConv\* [11], which utilized imbalanced vertex features where the central vertex had a dimension of 3 and its neighbors collectively contributed  $4 \times (K - 1)$  dimensions. Instead, we compute the mean of the neighboring edges to represent neighbor information, maintaining a dimension of 3, thereby ensuring a balanced feature representation between the center vertex and its neighborhood. This adjustment in SDConv (index G) leads to enhanced performance on both datasets when compared to SDConv\* (index C). Regarding the Gaussian random mapping matrix, our ablation study on making  $G$  trainable revealed no notable performance improvements. The primary function of  $G$  is to elevate the vertex feature from a lower to a higher dimension (from 6 to  $M$ ), and it appears that having a trainable matrix does not significantly impact performance and slightly increases the errors, likely due to the presence of a subsequent linear layer that processes the spectral feature.

By learning adaptive hierarchical mapping matrices for down-sampling and up-sampling in the context of hierarchical mesh representation learning (index H versus G), we can see a notable reduction in reconstruction errors on DFAUST while a slight increase for the COMA dataset (index H versus G). This disparity is attributed to the considerably greater variation in human body movements compared to facial expressions. Additionally, integrating a linear layer skip connection has further augmented performance. The absence of this skip connection (index D versus C) leads to heightened reconstruction errors in both datasets. This linear layer skip connection plays a crucial role in mitigating the gradient vanishing issue and emphasizes the significance of each vertex over its neighbors, thereby enhancing the overall effectiveness of the model.

Our examination extends to the impact of the basis size,  $B$ , for weighting matrices, with findings illustrated in Fig. 8. An increase in the basis size correlates with a decrease in reconstruction error, alongside a rise in the model’s size. Practically, this allows for a strategic selection of the basis size to navigate the trade-off between achieving higher reconstruction accuracy and maintaining a manageable model size, optimizing for both performance and efficiency.

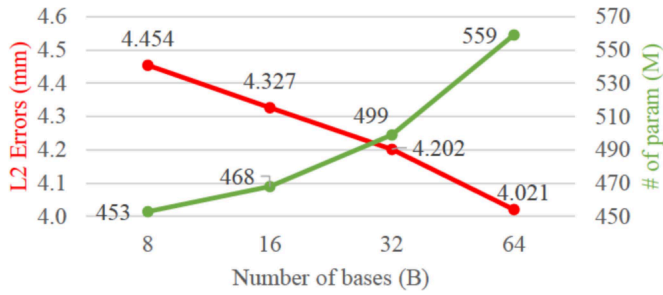


Fig. 8. Comparison of numbers of weighting matrix bases for SDConv\* [11] (no skip connection) on DFAUST when latent size  $d = 32$ . Blue denotes reconstruction error and orange model size.

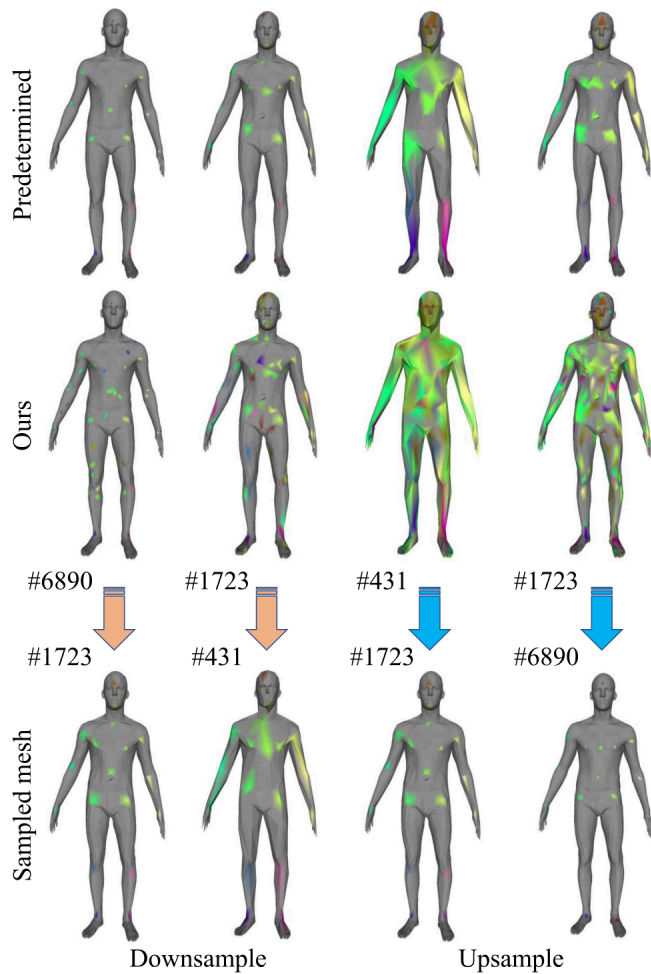


Fig. 9. Receptive fields of selected vertices at two levels of downsampling and upsampling on DFAUST dataset (best viewed in color). The last row (sampled mesh) shows the selected vertices with colors. The first row shows the receptive fields of the existing predetermined sampling method [13] at the neighboring level. The second row shows the receptive fields of our proposed adaptive sampling method at the neighboring level. ( $\#N_1 \Rightarrow \#N_2$ ) in the figure denotes the numbers of vertices before and after sampling.

#### F. Visualization of Matrices

To elucidate the advantages of employing learned hierarchical mapping matrices, we present visualization outcomes from the DFAUST dataset in Fig. 9. In these visualizations, the receptive

fields associated with selected vertices are highlighted in the same colors as the vertices themselves. This comparison reveals that the receptive fields determined by the predetermined sampling method [13] are confined to relatively small areas surrounding the selected vertices. Conversely, the receptive fields derived from our learned hierarchical mapping matrices extend over considerably larger regions, although they predominantly remain proximal to the selected vertices, demonstrating the enhanced spatial awareness and integration facilitated by our method.

Visualizations of the weighting matrices learned within our network on DFAUST are depicted in Fig. 10. The first two matrices showcased originate from the encoder, with the subsequent two matrices representing those from the decoder. Given that these weighting matrices are optimized without specific constraints, their values span both positive and negative ranges. It's observed that the learned weighting matrices tend to approximate the identity matrix, a tendency particularly pronounced in matrices derived from the encoder. This alignment with the identity matrix can be attributed to the initialization strategy of starting with an identity matrix. However, deviations from the identity matrix, informed by the local geometric structure of each vertex, play a pivotal role in the process of mesh representation learning, underscoring the adaptability and significance of these learned modifications.

#### G. Latent Feature Manipulation

We assess the representational capabilities of the learned latent features through three distinct operations: interpolation of latent features, mixing of local latent features, and transfer of deformations. These procedures serve to demonstrate the versatility and robustness of the latent features in capturing and manipulating the underlying geometric information of 3D shapes.

*Interpolation:* To demonstrate interpolation, we start with latent features  $z_1$  and  $z_2$  corresponding to distinct 3D shapes  $M_1$  and  $M_2$ . By linearly interpolating between these latent features, we generate a new latent feature  $z = \alpha z_1 + (1 - \alpha) z_2$ , where  $\alpha$  ranges between 0 and 1. The resulting shape from  $z$  is then reconstructed using the mesh decoder. Illustrations of interpolation results on the DFAUST, COMA, and MANO [27] datasets are provided in Fig. 11. These results reveal that interpolated shapes not only retain high visual quality but also smoothly transition across identities and poses (or expressions), showcasing the method's effectiveness in capturing nuanced shape variations.

Table VI provides quantitative validation of our interpolation quality across 100 randomly sampled shape pairs. The per-step error measurements demonstrate smooth and monotonic convergence toward the target shape, with errors decreasing consistently at each interpolation step for both datasets. For DFAUST, the error reduces from 242.37 mm to 58.12 mm across three steps, while COMA shows a proportionally similar reduction from 2.81 mm to 0.71 mm. The lower absolute errors in COMA reflect its focus on facial expressions with smaller geometric variations compared to DFAUST's full-body

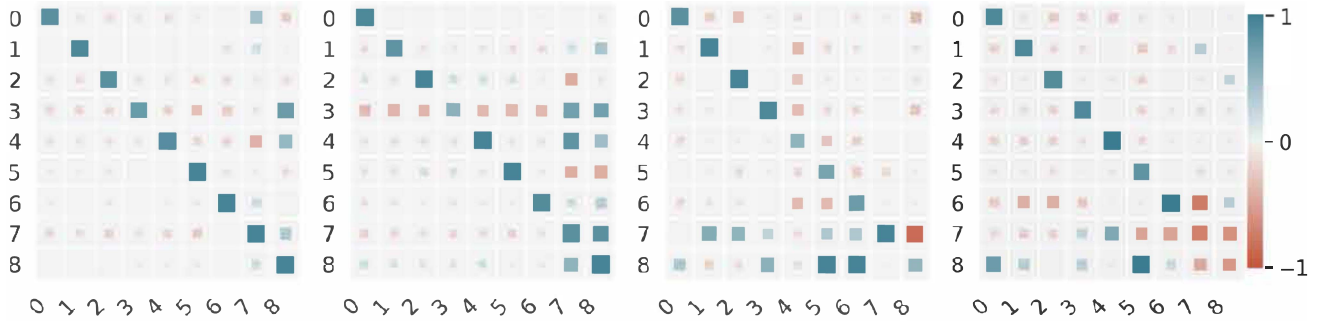


Fig. 10. Examples of weighting matrix bases at the finest level of the encoder and decoder on DFAUST dataset. The first two weighting matrices are from the encoder and the last two weighting matrices are from the decoder.

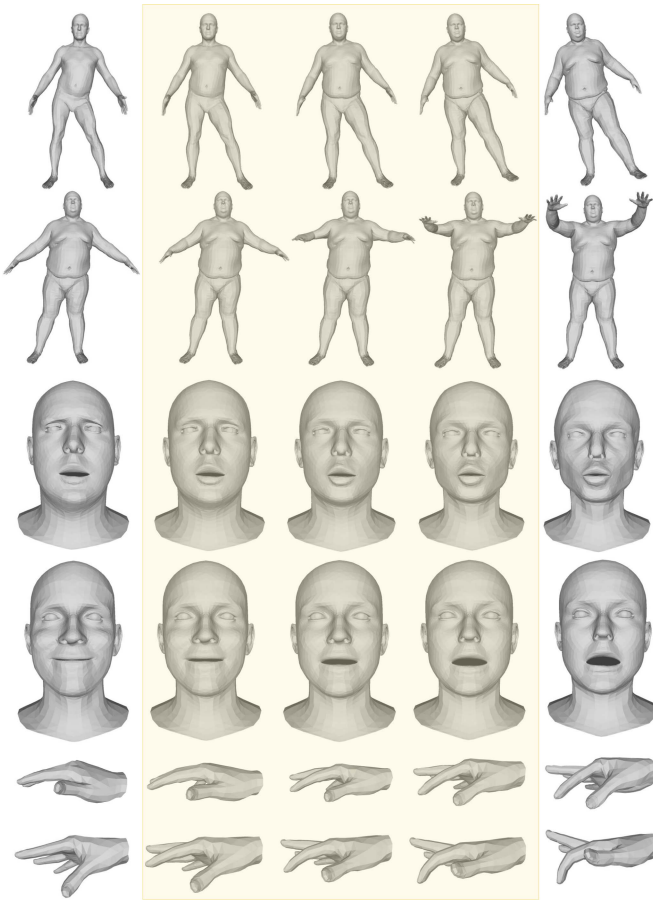


Fig. 11. Interpolations between different shapes on DFAUST and COMA dataset. The left-most and right-most shapes are input shapes, while the middle ones are interpolations.

poses. Critically, the low self-reconstruction errors (3.03 mm for DFAUST, 0.08 mm for COMA) confirm that linearly interpolated latent codes remain well within the learned manifold, producing geometrically valid shapes rather than artifacts. This validates that our latent space exhibits the smooth, continuous structure necessary for effective interpolation between arbitrary shapes.

**Mixture:** At the encoder's final layer, we acquire local latent features  $\mathbf{l}$  representing the shape's coarse topology, such as 27 vertices for bodies and 13 for hands. By taking local latent

TABLE VI  
QUANTITATIVE EVALUATION OF LATENT INTERPOLATION QUALITY OVER RANDOMLY SAMPLED 100 PAIR TEST MESHES. PER-STEP ERROR TO TARGET B MEASURES THE GEOMETRIC DISTANCE AT EACH OF 3 INTERPOLATION STEPS BETWEEN SOURCE AND TARGET SHAPES, DEMONSTRATING SMOOTH CONVERGENCE. SELF-RECONSTRUCTION ERROR VALIDATES THAT INTERPOLATED LATENT CODES REMAIN WITHIN THE LEARNED MANIFOLD.

| Metric                              | DFAUST | COMA |
|-------------------------------------|--------|------|
| Source A to Target B Error (mm)     | 242.37 | 2.81 |
| Step 1 to Target B Error (mm)       | 184.10 | 2.14 |
| Step 2 to Target B Error (mm)       | 120.95 | 1.43 |
| Step 3 to Target B Error (mm)       | 58.12  | 0.71 |
| Interpolation Self-Recon Error (mm) | 3.03   | 0.08 |

features  $\mathbf{l}_1$  and  $\mathbf{l}_2$  from two distinct 3D shapes  $M_1$  and  $M_2$ , we can substitute certain vertices of  $\mathbf{l}_1$  with their counterparts from  $\mathbf{l}_2$ . This process is formalized as:

$$\mathbf{l} = \mathbf{l}_1 \text{ and } \mathbf{l}[\mathcal{P}] = \mathbf{l}_2[\mathcal{P}], \quad (14)$$

where  $\mathcal{P}$  denotes the indices of vertices being replaced (for instance, the right arm as depicted in Fig. 12(a)). The novel shape generated from  $\mathbf{l}$  undergoes reconstruction through a fully-connected layer and mesh decoder. Fig. 12 showcases examples of mixing local latent features on both DFAUST and MANO datasets. The resultant new shape retains the original pose of  $M_1$ , except in the area specified by  $\mathcal{P}$ , which adopts the pose from  $M_2$ , demonstrating the technique's capability for precise pose manipulation within localized regions.

**Deformation Transfer:** It enables the transference of specific shape deformations across different identities. Consider  $\mathbf{z}_{p_0}$  and  $\mathbf{z}_{q_0}$  as latent features representing shapes from distinct identities yet sharing a similar pose or expression. If  $\mathbf{z}_{p_1}$  corresponds to a shape of the same identity as  $\mathbf{z}_{p_0}$  but exhibits a different pose or expression, the deformation representation can be captured as  $\mathbf{z}_D = \mathbf{z}_{p_1} - \mathbf{z}_{p_0}$ . This deformation  $\mathbf{z}_D$  can then be applied to  $\mathbf{z}_{q_0}$ , yielding  $\mathbf{z}_{q_1} = \mathbf{z}_{q_0} + \mathbf{z}_D$ , which represents the latent feature of the shape after deformation. The deformed shape, derived from  $\mathbf{z}_{q_1}$ , is subsequently constructed using the mesh decoder. Examples from the DFAUST and COMA datasets, shown in Fig. 13, illustrate the model's capability to accurately and effectively transfer deformations, maintaining high visual quality.

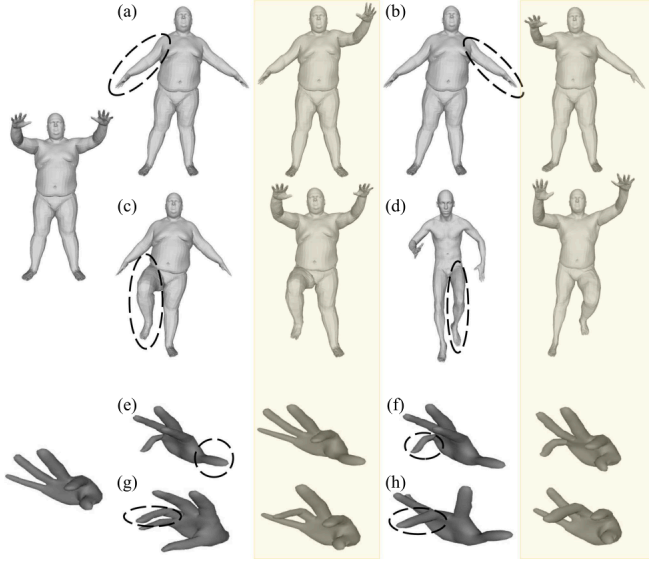


Fig. 12. Mixing local latent features from two shapes on DFAUST and MANO dataset. We replace the left-most shape's local latent feature that corresponds to the selected area (dashed ellipse) of another shape with the local latent feature of the selected area of that shape. The selected areas are the right arm in (a), left arm in (b), right leg in (c), left leg in (d), thumb in (e), little finger in (f), ring finger in (g), and middle finger in (h).

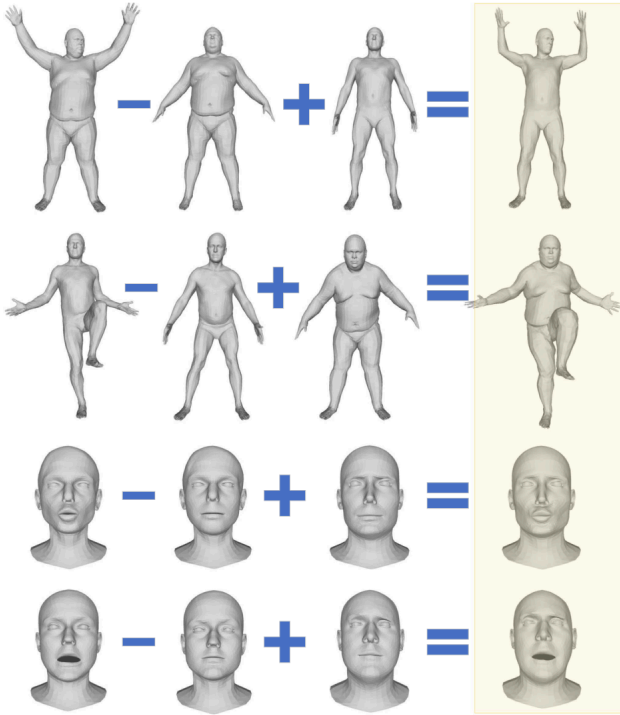


Fig. 13. Deformation transfer with arithmetic operations on latent representations on DFAUST and COMA dataset.

## VI. LIMITATIONS AND FUTURE DIRECTIONS

Our method is specifically designed for datasets with fixed topology and consistent vertex correspondence, such as COMA, DFAUST, and MANO. This is a deliberate architectural choice that provides significant advantages for this important problem

setting, rather than a limitation of the spectral dictionary framework.

*Design rationale and advantages:* The fixed-topology assumption enables three key benefits that distinguish our approach from point-based methods. First, our spectral dictionary construction exploits the fact that vertex  $i$  represents the same semantic location (e.g., nose tip, elbow) across all samples, allowing the learned bases  $\mathbf{D}$  and coefficients  $\mathbf{R}$  to capture meaningful geometric patterns rather than arbitrary spatial arrangements. Second, by pre-computing spectral features from consistent edge relationships (5), we achieve model complexity  $O(B \times K^2)$  that is independent of mesh resolution, where  $B$  denotes the dictionary size and  $K$  the neighborhood size. This contrasts sharply with methods that learn per-vertex weighting matrices directly (e.g., LSA-Conv with  $O(N \times K^2)$  parameters), enabling our approach to scale to high-resolution meshes without parameter explosion. Third, fixed topology facilitates shape correspondence tasks, latent space interpolation, and transfer learning across subjects, which are essential for 3D morphable models and character animation workflows.

*Scope of applicability:* These design choices make our method particularly well-suited for several important application domains: 3D morphable models for human faces and bodies, character animation and rigging, medical shape analysis where anatomical templates are standard, and industrial design workflows based on parametric templates. In these settings, which represent a substantial portion of mesh processing applications, consistent topology is not merely acceptable but often a design requirement that enables semantic correspondence.

*Current constraints:* The fixed-topology requirement means our approach cannot directly process raw scanned meshes with arbitrary triangulations or perform adaptive remeshing during processing. Applications such as reconstructing original mesh scans with varying connectivity or analyzing dynamic networks where topology evolves would require pre-processing steps or alternative approaches.

*Potential extensions:* While arbitrary topology support is not the focus of our current work, we envision several promising directions to expand the applicability of spectral dictionary learning. One practical approach involves template-based registration: fitting a standard mesh template to arbitrary input topology, applying our SDConv in template space, and transferring results back to the original mesh—a strategy commonly used in graphics pipelines (e.g., SMPL for bodies). Alternatively, learning dictionaries for local canonical patches rather than global vertex indices could enable transfer across meshes with similar local geometry but different global connectivity. For applications requiring true arbitrary topology handling without registration, dedicated point cloud architectures (e.g., Point-TransformerV3 [60]) that operate on unstructured point sets may be more appropriate, as they are designed specifically for that problem setting. A hybrid approach combining our spectral dictionary with point-based methods for irregular regions could also leverage the strengths of both paradigms.

Furthermore, the representational power of our spectral dictionary holds promise for high-fidelity mesh generation. By integrating with existing generative frameworks, our approach

could enable more detailed and controllable 3D content creation, pushing the boundaries of what is achievable in 3D modeling and generation while maintaining computational efficiency through our compact spectral representation.

## VII. CONCLUSION

We have developed a spectral dictionary-based convolutional neural network (SDConv) for mesh representation learning and demonstrate the performance on 3D shape generation tasks. We adopt the learnable adaptive-temperature softmax and spectral feature mapping functions to regress the coefficients of the weighting matrix bases (i.e., spectral dictionary) for each vertex. The weighting matrices are constructed by the linear combination of the bases in a weight-sharing manner to resample each vertex's neighbors. Then, a conventional CNN with anisotropic filters is applied. Furthermore, we introduce an adaptive sampling method to learn the hierarchical mapping matrices to improve the mesh representation power.

Compared with methods with either isotropic filters or pre-defined local coordinate systems, our SDConv and HSConv enjoy much higher representation power. Compared with methods that learn weighting matrices directly, our method achieves competitive or even better results with much smaller model sizes and can be used for applications that involve high-resolution 3D shapes and require high geometric details.

## REFERENCES

- [1] L. Tran and X. Liu, "Nonlinear 3D face morphable model," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7346–7355.
- [2] Y. Deng, J. Yang, S. Xu, D. Chen, Y. Jia, and X. Tong, "Accurate 3D face reconstruction with weakly-supervised learning: From single image to image set," in *Proc. IEEE/CVF Comput. Vis. Pattern Recognit. Workshops*, 2019, pp. 285–295.
- [3] Z. Gao, J. Zhang, Y. Guo, C. Ma, G. Zhai, and X. Yang, "Semi-supervised 3D face representation learning from unconstrained photo collections," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, 2020, pp. 1426–1435.
- [4] T. Groueix, M. Fisher, V. G. Kim, B. C. Russell, and M. Aubry, "3D-CODED: 3D correspondences by deep deformation," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 235–251.
- [5] O. Halimi, O. Litany, E. Rodola, A. M. Bronstein, and R. Kimmel, "Un-supervised learning of dense shape correspondence," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 4365–4374.
- [6] C. Cao, Y. Weng, S. Zhou, Y. Tong, and K. Zhou, "FaceWareHouse: A 3D facial expression database for visual computing," *IEEE Trans. Vis. Comput. Graphics*, vol. 20, no. 3, pp. 413–425, Mar. 2014.
- [7] F. Liu, R. Zhu, D. Zeng, Q. Zhao, and X. Liu, "Disentangling features in 3D face shapes for joint face reconstruction and recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 5216–5225.
- [8] M. onlić, T. Petković, S. Peharec, F. Berryman, and T. Pribanić, "On the segmentation of scanned 3D human body models," in *Proc. 8th Int. Sci. Conf. Kinesiol.*, 2017, Art. no. 694.
- [9] E. Kalogerakis, A. Hertzmann, and K. Singh, "Learning 3D mesh segmentation and labeling," in *Proc. ACM SIGGRAPH 2010 Papers*, 2010, pp. 1–12.
- [10] C. Cao, Q. Hou, and K. Zhou, "Displaced dynamic expression regression for real-time facial tracking and animation," *ACM Trans. Graph.*, vol. 33, no. 4, Jul. 2014, Art. no. 43.
- [11] Z. Gao, J. Yan, G. Zhai, and X. Yang, "Learning spectral dictionary for local representation of mesh," in *Proc. 13th Int. Joint Conf. Artif. Intell.*, 2021, pp. 685–692.
- [12] Z. Gao, J. Yan, G. Zhai, J. Zhang, Y. Yang, and X. Yang, "Learning local neighboring structure for robust 3D shape representation," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 1397–1405.
- [13] A. Ranjan, T. Bolkart, S. Sanyal, and M. J. Black, "Generating 3D faces using convolutional mesh autoencoders," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 725–741.
- [14] G. Bouritsas, S. Bokhnyak, S. Ploumpis, M. Bronstein, and S. Zafeiriou, "Neural 3D morphable models: Spiral convolutional networks for 3D shape representation learning and generation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 7212–7221.
- [15] S. Gong, L. Chen, M. Bronstein, and S. Zafeiriou, "SpiralNet++: A fast and highly efficient mesh convolution operator," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops*, 2019, pp. 4141–4148.
- [16] F. Babiloni, M. Maggioni, T. Tanay, J. Deng, A. Leonardis, and S. Zafeiriou, "Adaptive spiral layers for efficient 3D representation learning on meshes," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 14620–14631.
- [17] Y. Zhou et al., "Fully convolutional mesh autoencoder using efficient spatially varying kernels," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2020, pp. 9251–9262.
- [18] N. Verma, E. Boyer, and J. Verbeek, "FeaStNet: Feature-steered graph convolutions for 3D shape analysis," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 2598–2606.
- [19] F. Bogo, J. Romero, G. Pons-Moll, and M. J. Black, "Dynamic FAUST: Registering human bodies in motion," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 5573–5582.
- [20] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2016, pp. 3844–3852.
- [21] Z. Chen and T.-K. Kim, "Learning feature aggregation for deep 3D morphable models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 13164–13173.
- [22] Z. Gao, "Learning continuous mesh representation with spherical implicit surface," in *Proc. IEEE 17th Int. Conf. Autom. Face Gesture Recognit.*, 2023, pp. 1–8.
- [23] Y. Li, R. Bu, M. Sun, W. Wu, X. Di, and B. Chen, "PointCNN: Convolution on  $\mathcal{X}$ -transformed points," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2018, pp. 828–838.
- [24] C. Zimmermann, D. Ceylan, J. Yang, B. Russell, M. Argus, and T. Brox, "FreiHAND: A dataset for markerless capture of hand pose and shape from single RGB images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 813–822.
- [25] P. Paysan, R. Knothe, B. Amberg, S. Romdhani, and T. Vetter, "A 3D face model for pose and illumination invariant face recognition," in *Proc. 6th IEEE Int. Conf. Adv. Video Signal Based Surveill.*, 2009, pp. 296–301.
- [26] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black, "SMPL: A skinned multi-person linear model," *ACM Trans. Graph.*, vol. 34, no. 6, Oct. 2015, Art. no. 248.
- [27] J. Romero, D. Tzionas, and M. J. Black, "Embodied hands: Modeling and capturing hands and bodies together," *ACM Trans. Graph.*, vol. 36, no. 6, Nov. 2017, Art. no. 245.
- [28] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Representations*, 2017, pp. 1–14.
- [29] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 1025–1035.
- [30] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Representations*, 2018, pp. 1–12.
- [31] F. Monti, D. Boscaini, J. Masci, E. Rodola, J. Svoboda, and M. M. Bronstein, "Geometric deep learning on graphs and manifolds using mixture model CNNs," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 5425–5434.
- [32] Q. Huang, X. Huang, B. Sun, Z. Zhang, J. Jiang, and C. Bajaj, "ARAPReg: An as-rigid-as possible regularization loss for learning deformable shape generators," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 5815–5825.
- [33] H. Yang, B. Sun, L. Chen, A. Pavel, and Q. Huang, "GeoLatent: A geometric approach to latent space design for deformable shape generators," *ACM Trans. Graph.*, vol. 42, no. 6, pp. 1–20, 2023.
- [34] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "PointNet: Deep learning on point sets for 3D classification and segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 77–85.
- [35] J. Li, B. M. Chen, and G. H. Lee, "SO-Net: Self-organizing network for point cloud analysis," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 9397–9406.

- [36] H. Zhao, L. Jiang, C.-W. Fu, and J. Jia, "PointWeb: Enhancing local neighborhood features for point cloud processing," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 5560–5568.
- [37] Z. Zhang, B.-S. Hua, and S.-K. Yeung, "ShellNet: Efficient point cloud convolutional neural networks using concentric shells statistics," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2019, pp. 1607–1616.
- [38] Y. Wang, Y. Sun, Z. Liu, S. E. Sarma, M. M. Bronstein, and J. M. Solomon, "Dynamic graph CNN for learning on point clouds," *ACM Trans. Graph.*, vol. 38, no. 5, Oct. 2019, Art. no. 146.
- [39] Y. Liu, B. Fan, S. Xiang, and C. Pan, "Relation-shape convolutional neural network for point cloud analysis," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 8895–8904.
- [40] H. Thomas, C. R. Qi, J.-E. Deschaud, B. Marcotegui, F. Goulette, and L. J. Guibas, "KPConv: Flexible and deformable convolution for point clouds," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 6410–6419.
- [41] F. Bogo, J. Romero, M. Loper, and M. J. Black, "FAUST: Dataset and evaluation for 3D mesh registration," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2014, pp. 3794–3801.
- [42] A. Tewari et al., "Self-supervised multi-level face model learning for monocular reconstruction at over 250 Hz," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 2549–2559.
- [43] A. T. Tran, T. Hassner, I. Masi, E. Paz, Y. Nirkin, and G. Medioni, "Extreme 3D face reconstruction: Seeing through occlusions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 3935–3944.
- [44] Y. Guo, J. Zhang, J. Cai, B. Jiang, and J. Zheng, "CNN-based real-time dense face reconstruction with inverse-rendered photo-realistic face images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 6, pp. 1294–1307, Jun. 2019.
- [45] L. Tran, F. Liu, and X. Liu, "Towards high-fidelity nonlinear 3D face morphable model," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 1126–1135.
- [46] D. Kulon, H. Wang, R. A. Güler, M. M. Bronstein, and S. Zafeiriou, "Single image 3D hand reconstruction with mesh convolutions," in *Proc. Brit. Mach. Vis. Conf.*, 2019, Art. no. 45.
- [47] K. Lin, L. Wang, and Z. Liu, "End-to-end human pose and mesh reconstruction with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 1954–1963.
- [48] A. Vaswani et al., "Attention is all you need," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [49] Y. Liu, Q. Chen, W. Chen, and I. Wassell, "Dictionary learning inspired deep network for scene recognition," in *Proc. AAAI Conf. Artif. Intell.*, 2018, Art. no. 879.
- [50] S. Mahdizadehaghdam, A. Panahi, H. Krim, and L. Dai, "Deep dictionary learning: A parametric network approach," *IEEE Trans. Image Process.*, vol. 28, no. 10, pp. 4790–4802, Oct. 2019.
- [51] D.-A. Clevert, T. Unterthiner, and S. Hochreiter, "Fast and accurate deep network learning by exponential linear units (ELUs)," 2015, *arXiv:1511.07289*.
- [52] A. Rahimi and B. Recht, "Random features for large-scale kernel machines," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2007, pp. 1177–1184.
- [53] Z. Gao, G. Zhai, J. Yan, and X. Yang, "Permutation matters: Anisotropic convolutional layer for learning on point clouds," 2020, *arXiv:2005.13135v2*.
- [54] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
- [55] X. Zhou, M. Zhu, G. Pavlakos, S. Leonardos, K. G. Derpanis, and K. Daniilidis, "MonoCap: Monocular human motion capture using a CNN coupled with a geometric prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 4, pp. 901–914, Apr. 2019.
- [56] J. C. Gower, "Generalized procrustes analysis," *Psychometrika*, vol. 40, no. 1, pp. 33–51, 1975.
- [57] M. Wei et al., "AGConv: Adaptive graph convolution on 3D point clouds," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 9374–9392, Aug. 2023.
- [58] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [59] I. T. Jolliffe, *Principal Component Analysis and Factor Analysis*. New York, NY, USA: Springer, 1986, pp. 115–128. [Online]. Available: [https://doi.org/10.1007/978-1-4757-1904-8\\_7](https://doi.org/10.1007/978-1-4757-1904-8_7)
- [60] X. Wu et al., "Point transformer V3: Simpler faster stronger," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 4840–4851.



**Zhongpai Gao** (Member, IEEE) received the BSc degree in electronic and information engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2013, and the PhD degree from Shanghai Jiao Tong University, Shanghai, China, in 2018. He was a visiting PhD student with the Schepens Eye Research Institute, Harvard Medical School, Boston, USA, from 2016 to 2018. He was a postdoctoral fellow with Shanghai Jiao Tong University from 2018 to 2021. Currently, he is a senior expert research scientist with UII America, Boston, MA, USA. His research interests include 3D computer vision and AI for healthcare. He serves as area chair for CVPR, ICLR, WACV, BMVC, etc. He has published more than 20 publications in top venues.



**Junchi Yan** (Senior Member, IEEE) received the PhD degree in electronic engineering from Shanghai Jiao Tong University. He is a full professor (tenured) with the Department of Computer Science and Engineering, Shanghai Jiao Tong University, Shanghai, China. Before that, he was a senior research staff member and principal scientist with IBM Research where he started his career since April 2011. His main research interest is machine learning and its intersection with other areas e.g. operations research and quantum computing. He serves as associate editor of the *Pattern Recognition Journal* and *ACM Transactions on Probabilistic Machine Learning*, and (senior) area chair/senior PC for ICML, ICLR, NeurIPS, SIGKDD, AAAI, IJCAI, CVPR, etc. He has published more than 200 publications in top venues.



**Tianyu Luan** received the BS degree in applied physics from the University of Science and Technology of China, Hefei Anhui, China, in 2013, and the MEng degree in electronical and telecommunication engineering from Tsinghua University, Beijing, China, in 2017. He is currently working toward the PhD degree with the Computer Science & Engineering Department, State University of New York at Buffalo, Buffalo NY, United States, advised by Prof. Junsong Yuan. His research interest lies in 3D human perception prior & 3D generation.



**Guangtao Zhai** (Fellow, IEEE) received the BE and ME degrees from Shandong University, Shandong, China, in 2001 and 2004, respectively, and the PhD degree from Shanghai Jiao Tong University, Shanghai, China, in 2009. He is currently a research professor with the Institute of Image Communication and Information Processing, Shanghai Jiao Tong University. He was the recipient of the Award of National Excellent PhD thesis from the Ministry of Education of China in 2012. From 2008 to 2009, he was a visiting student with the Department of Electrical and Computer Engineering, McMaster University, Hamilton, ON, Canada, where he was a postdoctoral fellow from 2010 to 2012. From 2012 to 2013, he was a Humboldt research fellow with the Institute of Multimedia Communication and Signal Processing, Friedrich Alexander University of Erlangen-Nuremberg, Erlangen, Germany. His research interests include multimedia signal processing and perceptual signal processing. He was the recipient of the Award of National Excellent PhD thesis from the Ministry of Education of China, in 2012.



**Xiaokang Yang** (Fellow, IEEE) received the BS degree from Xiamen University, in 1994, the MS degree from the Chinese Academy of Sciences, in 1997, and the PhD degree from Shanghai Jiao Tong University, in 2000. He is currently a distinguished professor with the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China. His research interests include visual signal processing and communication, media analysis and retrieval, and pattern recognition. He serves as an associate editor of *IEEE Transactions on Multimedia*, *IEEE Signal Processing Letters*.